

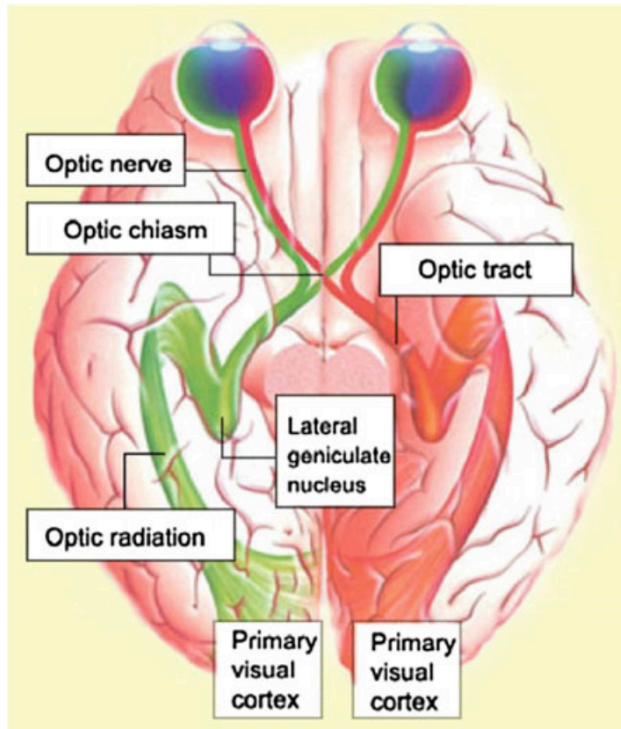
Neurogeometry

Part IV: Neurogeometric modeling of visual perception

Retino-geniculo-cortical pathway

Focus:

- Human, primates, cats (not all mammals, e. g., not mice!)
- First step of visual perception in the cortex — Primary visual cortex



Retina



Optic nerve

Optic chiasm: cross-point that organizes optic nerves properly for coordinated binocular vision



Optic tract

Lateral geniculate nucleus (LGN): thalamic relay transmitting visual information to the primary visual cortex



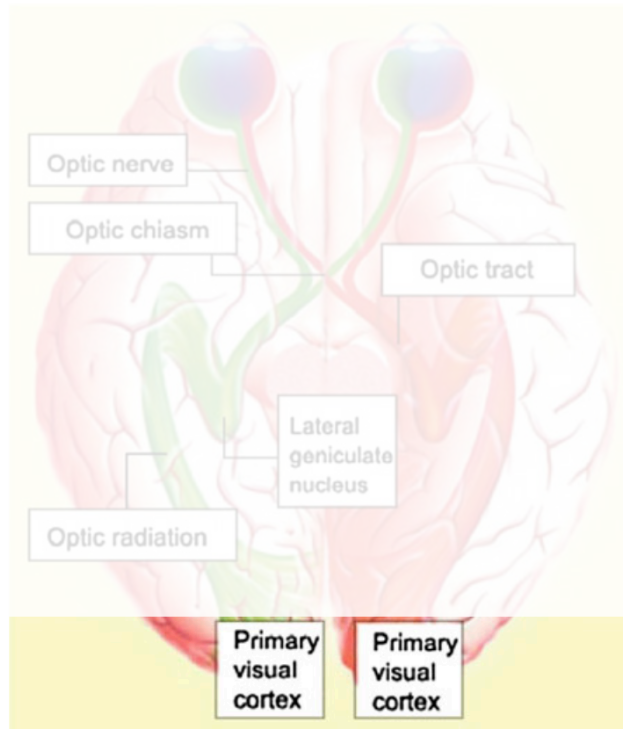
Optic radiation

Primary visual cortex

Retino-geniculo-cortical pathway

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- Human, primates, cats (not all mammals, e. g., not mice!)
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Retina

↓
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Optic tract

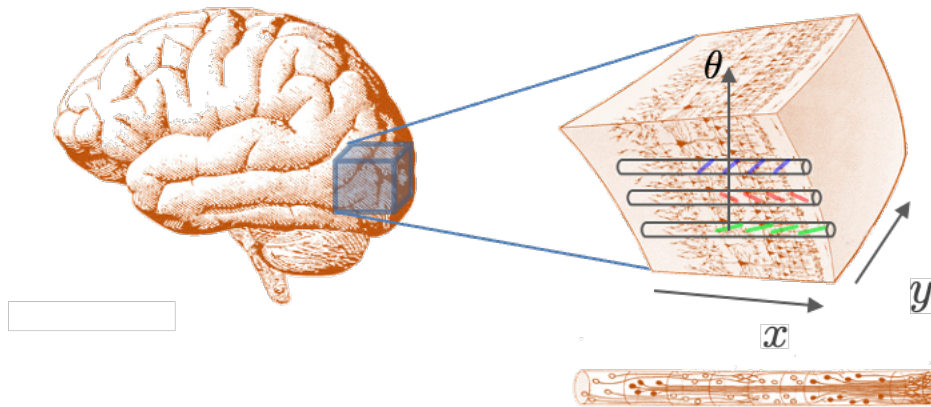
Lateral geniculate nucleus (LGN): thalamic relay transmitting visual information to the primary visual cortex

↓
Optic radiation

Primary visual cortex

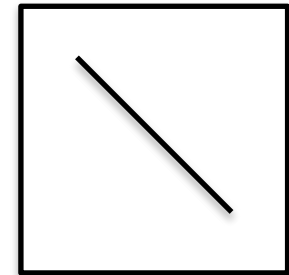
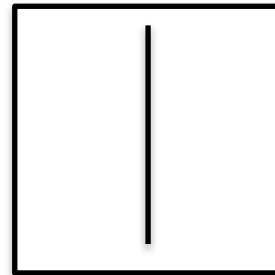
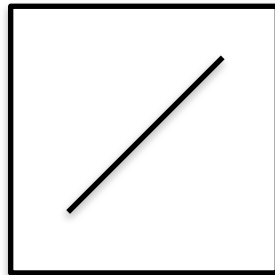
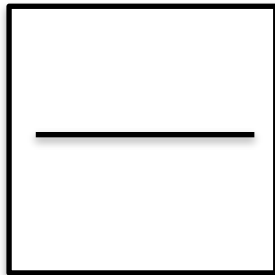
Petitot, Neurogéométrie de la vision,
2008

Primary visual cortex (V1)

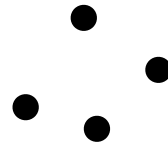
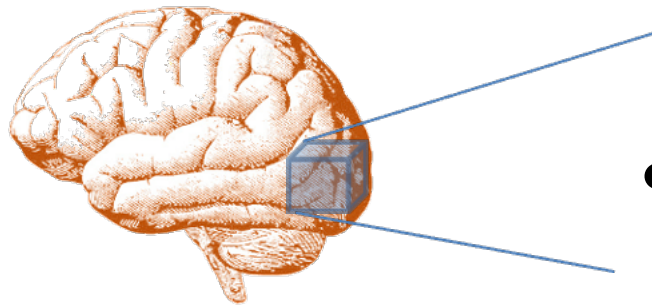


Simple cells

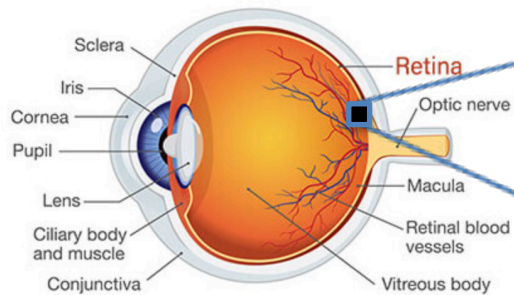
- Located in the occipital lobe (backside)
- Responsible for first step processing of the visual information transmitted from retina
- Sensitive to visual features of the input, **in particular to orientation**



Simple cells, receptive fields, receptive profiles



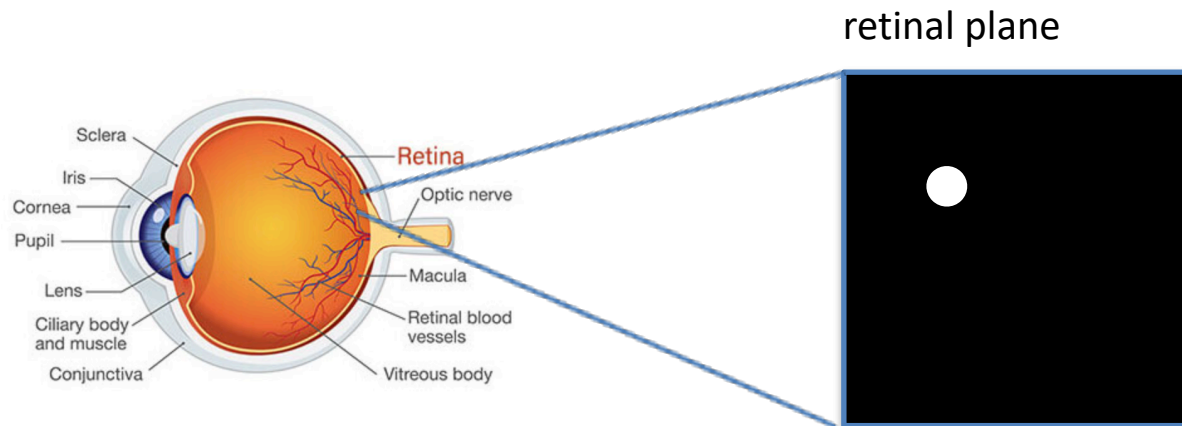
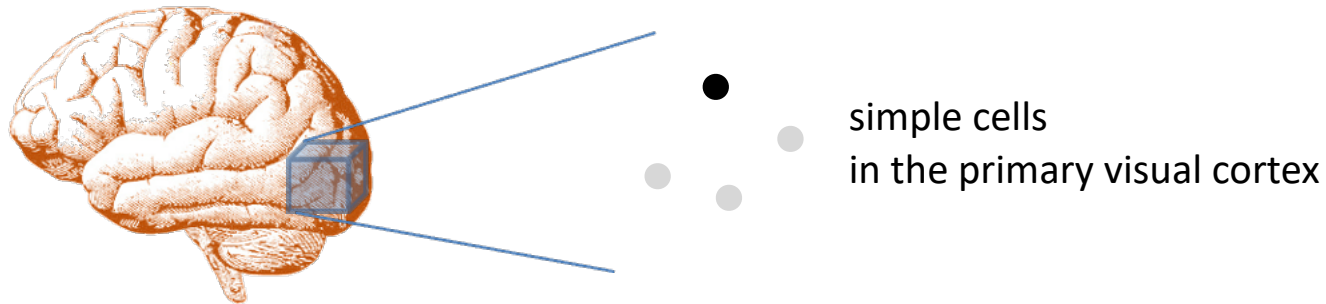
simple cells
in the primary visual cortex



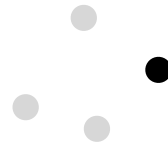
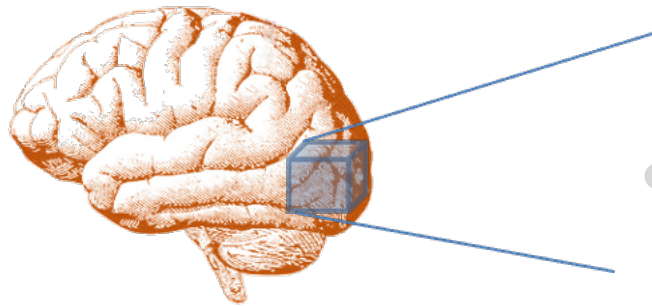
retinal plane



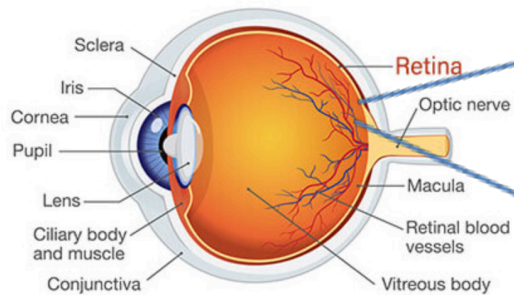
Simple cells, receptive fields, receptive profiles



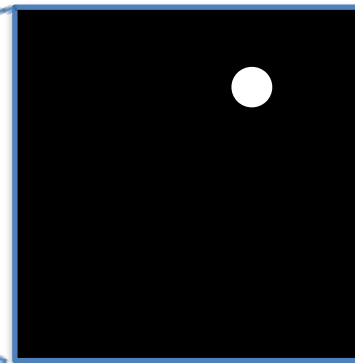
Simple cells, receptive fields, receptive profiles



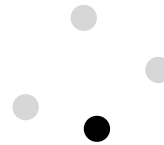
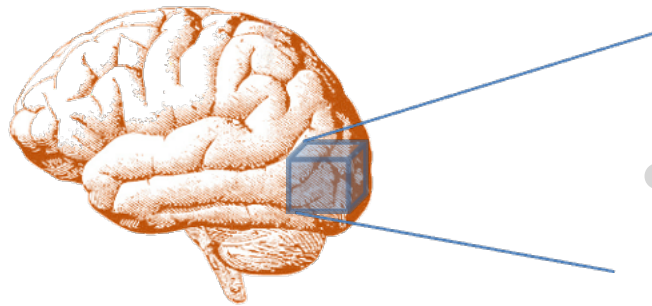
simple cells
in the primary visual cortex



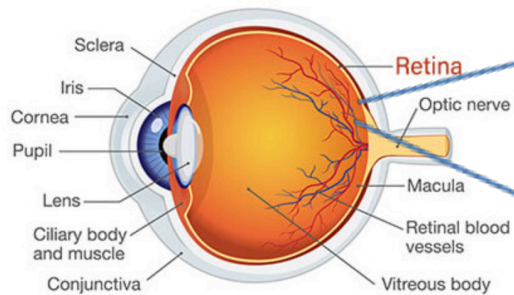
retinal plane



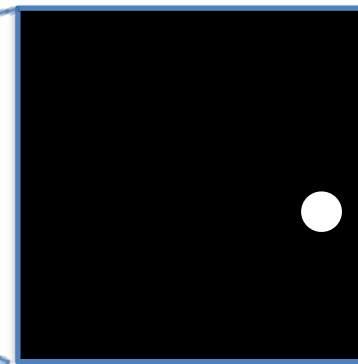
Simple cells, receptive fields, receptive profiles



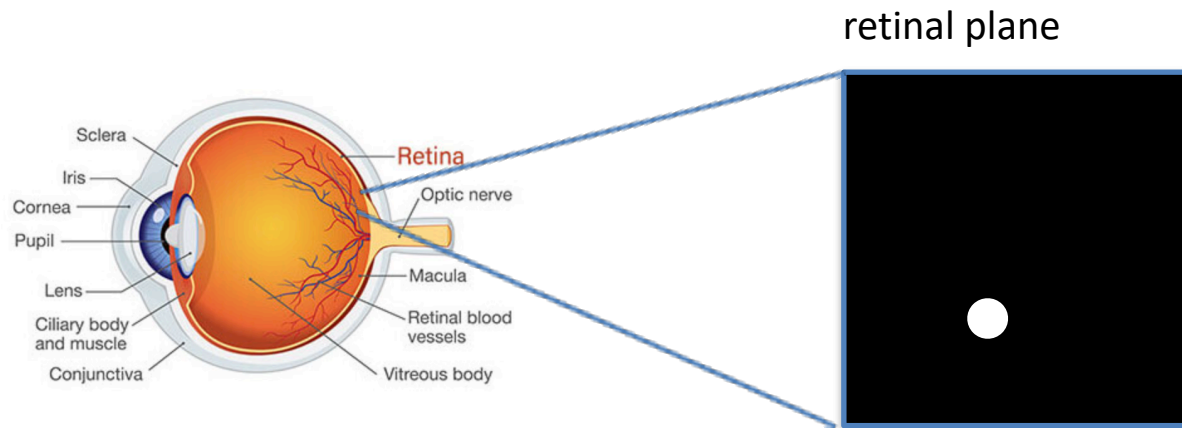
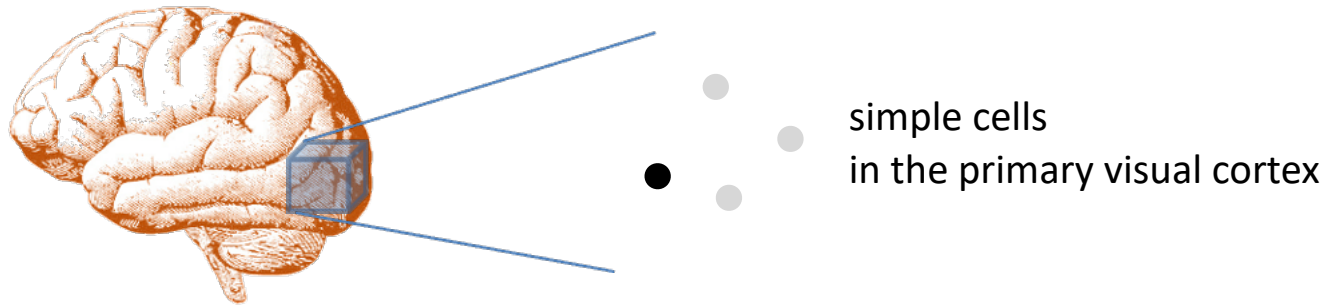
simple cells
in the primary visual cortex



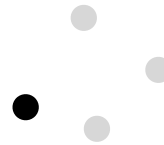
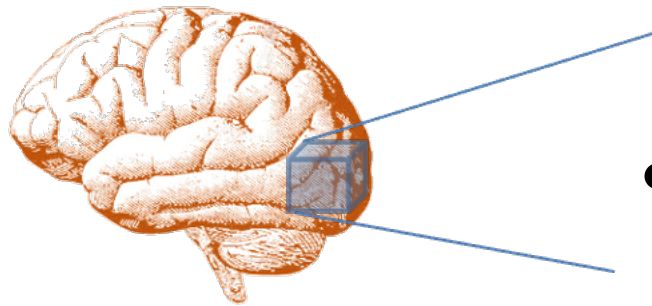
retinal plane



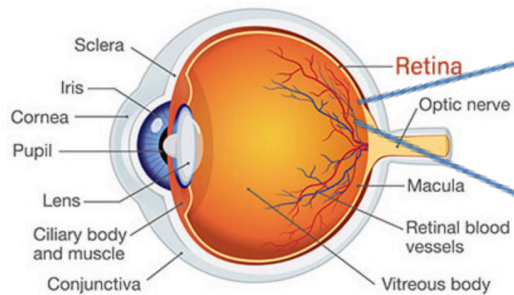
Simple cells, receptive fields, receptive profiles



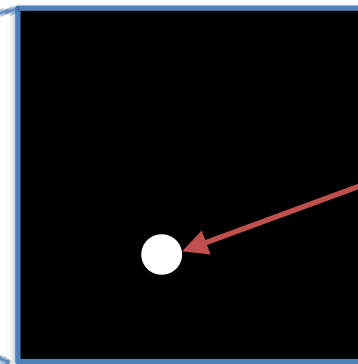
Simple cells, receptive fields, receptive profiles



simple cells
in the primary visual cortex

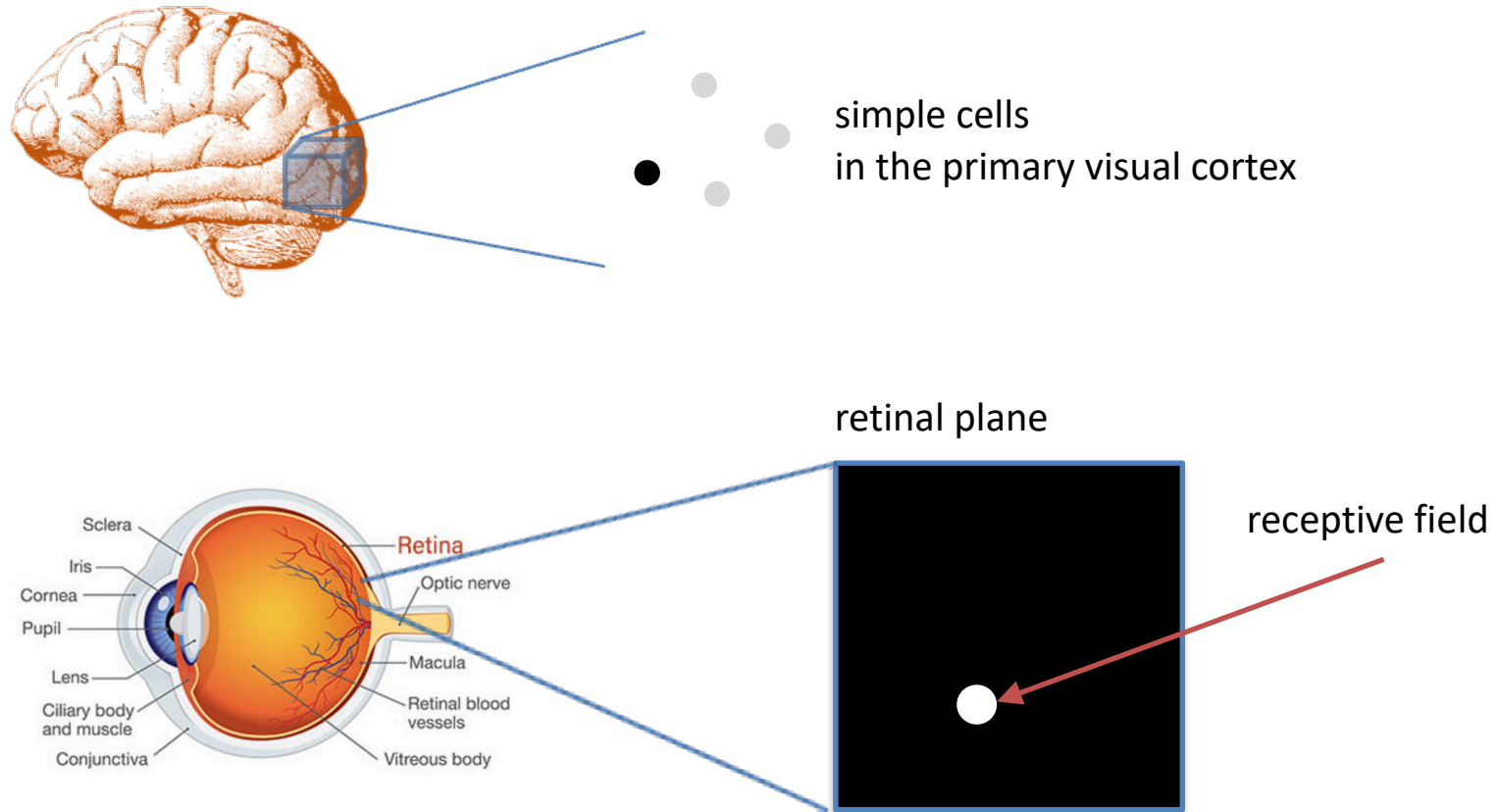


retinal plane



receptive field

Simple cells, receptive fields, receptive profiles

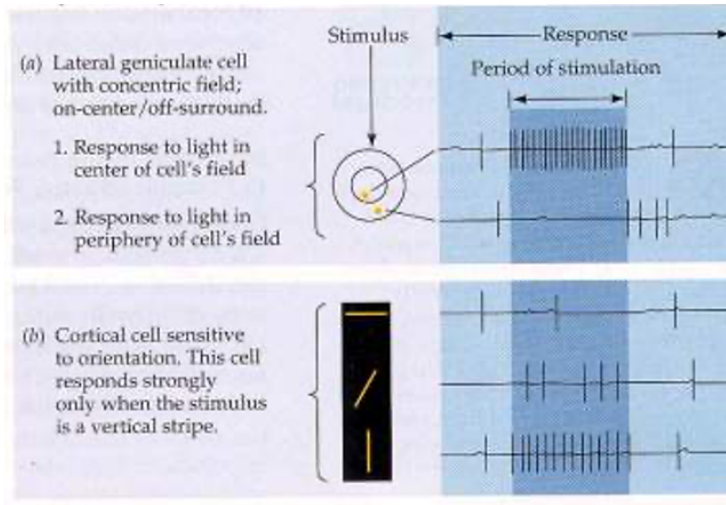


Receptive field:

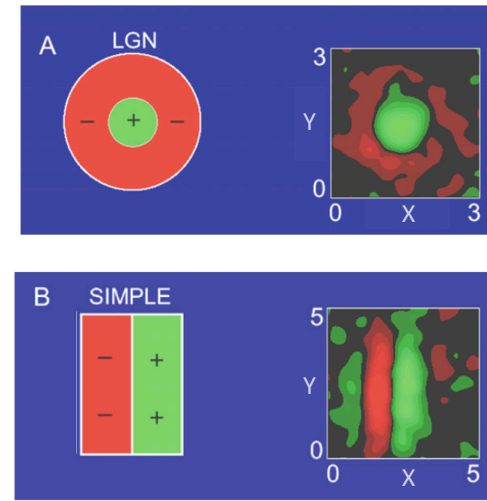
- Retinal zone to which the simple cell is connected through retino-geniculo-cortical pathways.
- Simple cell produces spikes (gives response) when its receptive field is stimulated.

Simple cells, receptive fields, receptive profiles

Response of a light sensitive LGN



Response of a V1 simple cell



From http://jeanpetitot.com/Woodruff-Pak_vision.ppt

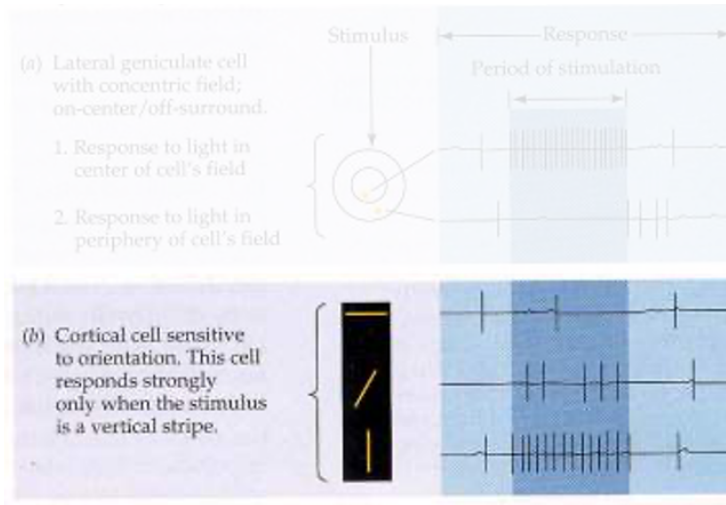
Petitot, Neurogéométrie de la vision, 2008

Receptive profile:

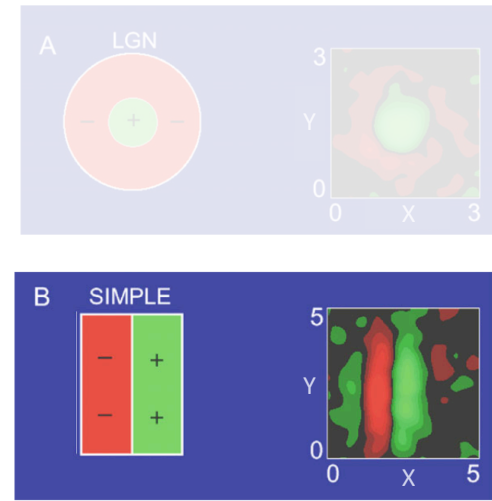
- Impulse response function of a neuron in the visual system.
- Elongated in simple cells → Orientation sensitivity

Simple cells, receptive fields, receptive profiles

Response of a light sensitive LGN



Response of a V1 simple cell



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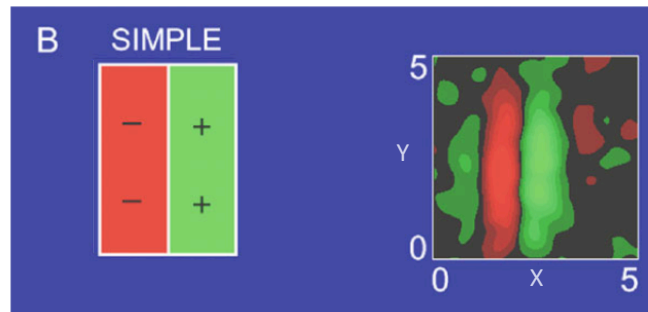
Petitot, Neurogéométrie de la vision, 2008

Receptive profile:

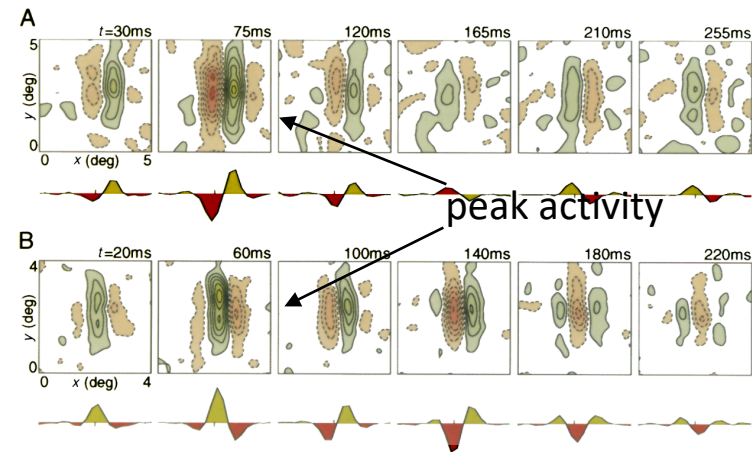
- Impulse response function of a neuron in the visual system.
- Elongated in simple cells → Orientation sensitivity

Modeling simple cell receptive profiles

Receptive profile (experimental)

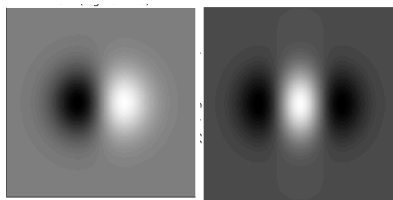


Petitot, Neurogéométrie de la vision, 2008

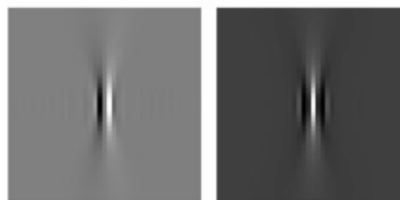


DeAngelis et al., Trends Neurosci., 1995

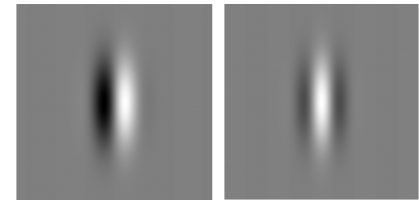
Receptive profile (mathematical model)



Gaussian derivatives



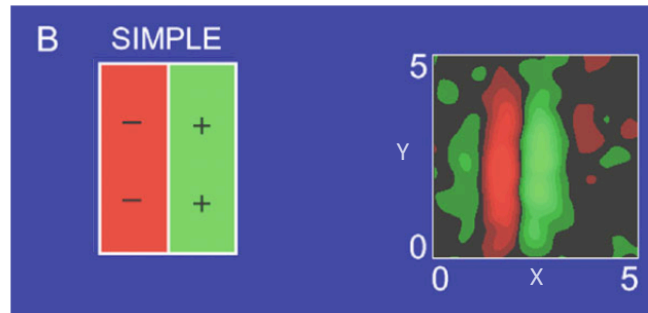
Gabor kernels



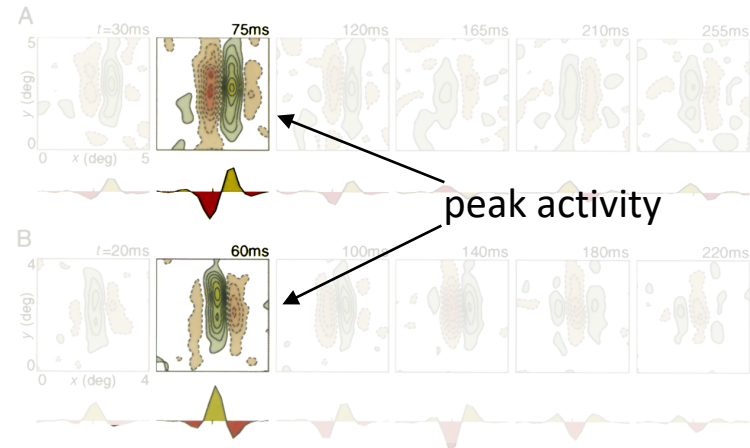
Gabor functions

Modeling simple cell receptive profiles

Receptive profile (experimental)

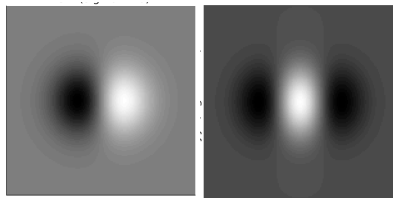


Petitot, Neurogéométrie de la vision, 2008

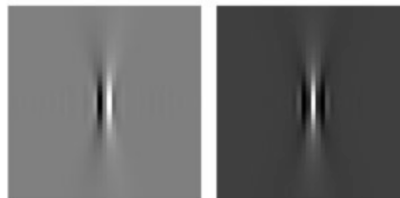


DeAngelis et al., Trends Neurosci., 1995

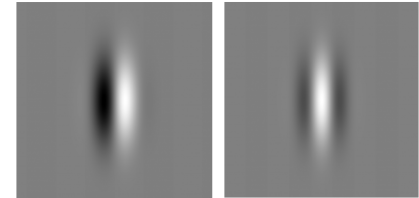
Receptive profile (mathematical model)



Gaussian derivatives



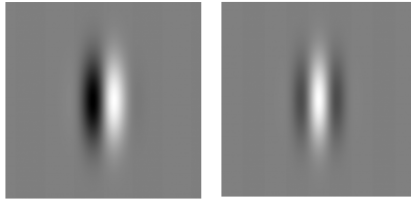
Gabor kernels



Gabor functions

Modeling simple cell receptive profiles

odd (sin)
component even (cos)
component



Gabor functions

$$(\exp(i x) = \cos x + i \sin x)$$



$$\Psi_{(x_0, y_0, \theta_0)}(x, y) = \frac{1}{2 \pi \sigma^2} \exp\left(-\frac{\xi^2 + \eta^2}{\sigma^2}\right) \exp\left(i \frac{\eta}{\sigma}\right)$$



Normalization
constant



Localizing Gaussian
window



Wave content

x_0, y_0, θ_0 : coordinate parameters

σ : scale

$$\xi = (x - x_0) \cos \theta_0 + (y - y_0) \sin \theta_0$$

$$\eta = -(x - x_0) \sin \theta_0 + (y - y_0) \cos \theta_0$$

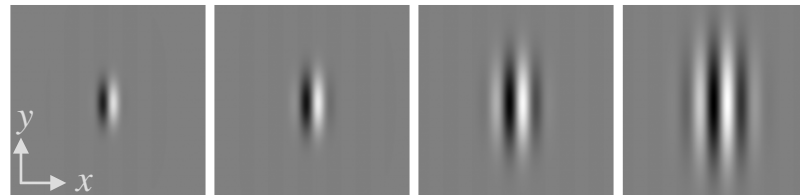
$$\begin{pmatrix} \xi \\ \eta \end{pmatrix} = R_{\theta_0}^{-1} \begin{pmatrix} x - x_0 \\ y - y_0 \end{pmatrix}$$

$$R_{\theta_0} = \begin{pmatrix} \cos \theta_0 & -\sin \theta_0 \\ \sin \theta_0 & \cos \theta_0 \end{pmatrix}$$

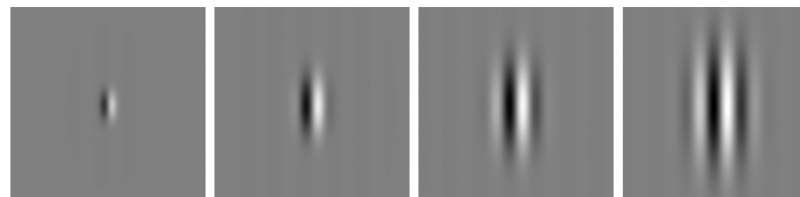
σ increases



odd (sin)
component

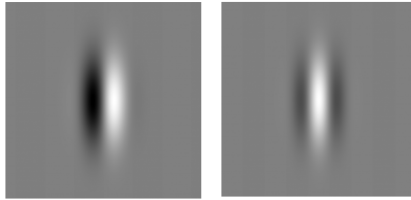


even (cos)
component



Modeling simple cell receptive profiles

odd (sin)
component even (cos)
component



Gabor functions

$$\Psi_{(x_0, y_0, \theta_0)}(x, y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{\xi^2 + \eta^2}{\sigma^2}\right) \exp\left(i\frac{\eta}{\sigma}\right)$$

$\uparrow$ $\uparrow$ $\uparrow$
 Normalization Localizing Gaussian Wave content
 constant window

$(\exp(i x) = \cos x + i \sin x)$
 $\downarrow$

x_0, y_0, θ_0 : coordinate parameters

σ : scale

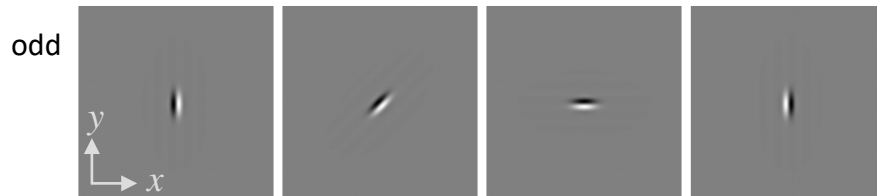
$$\xi = (x - x_0) \cos \theta_0 + (y - y_0) \sin \theta_0$$

$$\eta = -(x - x_0) \sin \theta_0 + (y - y_0) \cos \theta_0$$

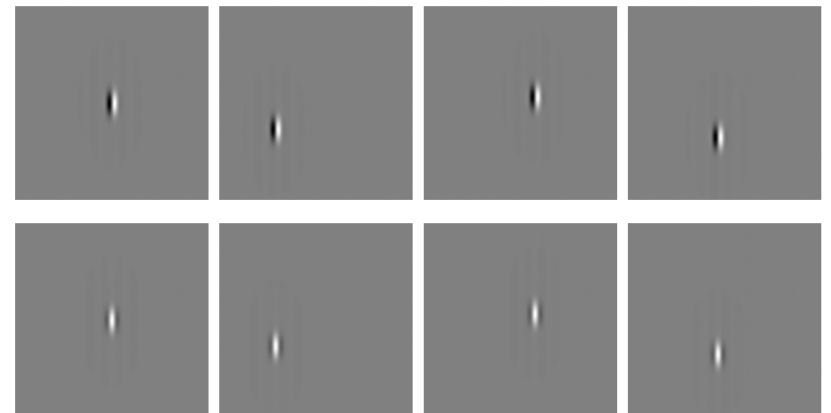
$$\begin{pmatrix} \xi \\ \eta \end{pmatrix} = R_{\theta_0}^{-1} \begin{pmatrix} x - x_0 \\ y - y_0 \end{pmatrix}$$

$$R_{\theta_0} = \begin{pmatrix} \cos \theta_0 & -\sin \theta_0 \\ \sin \theta_0 & \cos \theta_0 \end{pmatrix}$$

$\theta_0 = 0$ $\theta_0 = \pi/4$ $\theta_0 = \pi/2$ $\theta_0 = \pi$



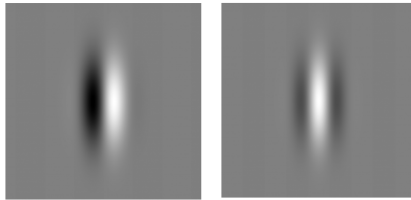
(x_0, y_0) varies



Modeling simple cell receptive profiles

odd (sin)
component

even (cos)
component



Gabor functions

$$(\exp(i x) = \cos x + i \sin x)$$



$$\Psi_{(x_0, y_0, \theta_0)}(x, y) = \frac{1}{2 \pi \sigma^2} \exp\left(-\frac{\xi^2 + \eta^2}{\sigma^2}\right) \exp\left(i \frac{\eta}{\sigma}\right)$$



Normalization
constant



Localizing Gaussian
window



Wave content

x_0, y_0, θ_0 : coordinate parameters

σ : scale

$$\xi = (x - x_0) \cos \theta_0 + (y - y_0) \sin \theta_0$$

$$\eta = -(x - x_0) \sin \theta_0 + (y - y_0) \cos \theta_0$$

$$\begin{pmatrix} \xi \\ \eta \end{pmatrix} = R_{\theta_0}^{-1} \begin{pmatrix} x - x_0 \\ y - y_0 \end{pmatrix}$$

$$R_{\theta_0} = \begin{pmatrix} \cos \theta_0 & -\sin \theta_0 \\ \sin \theta_0 & \cos \theta_0 \end{pmatrix}$$

Gabor functions can be generated from a reference Gabor function: Gabor *mother* function.

$\theta_0 = 0$

$\theta_0 = \pi/4$

$\theta_0 = \pi/2$

$\theta_0 = \pi$

(x_0, y_0) varies

odd

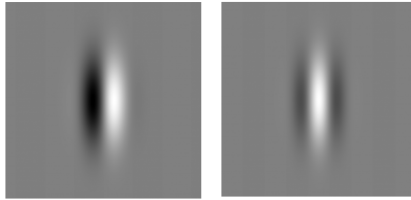


even



Modeling simple cell receptive profiles

odd (sin)
component even (cos)
component



Gabor functions

$$(\exp(i x) = \cos x + i \sin x)$$



$$\Psi_{(x_0, y_0, \theta_0)}(x, y) = \frac{1}{2 \pi \sigma^2} \exp\left(-\frac{\xi^2 + \eta^2}{\sigma^2}\right) \exp\left(i \frac{\eta}{\sigma}\right)$$



Normalization
constant



Localizing Gaussian
window



Wave content

$$\begin{aligned} \xi &= (x - x_0) \cos \theta_0 + (y - y_0) \sin \theta_0 \\ \eta &= -(x - x_0) \sin \theta_0 + (y - y_0) \cos \theta_0 \end{aligned}$$

$$\begin{pmatrix} \xi \\ \eta \end{pmatrix} = R_{\theta_0}^{-1} \begin{pmatrix} x - x_0 \\ y - y_0 \end{pmatrix} \quad R_{\theta_0} = \begin{pmatrix} \cos \theta_0 & -\sin \theta_0 \\ \sin \theta_0 & \cos \theta_0 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \underbrace{\begin{pmatrix} x_0 \\ y_0 \end{pmatrix}}_{\text{translation}} + \underbrace{R_{\theta_0}}_{\text{rotation}} \begin{pmatrix} \xi \\ \eta \end{pmatrix}$$



Let us write this in terms of a transfer operator:

$$(x, y) = A_{(x_0, y_0, \theta_0)}(\xi, \eta).$$

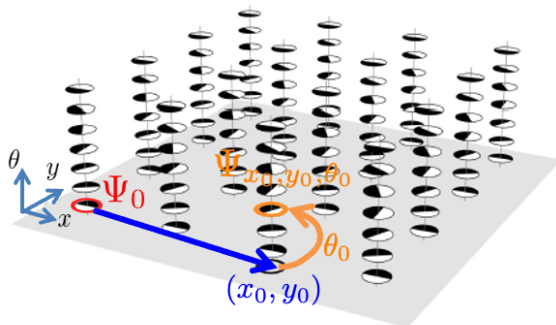
So we can write (ξ, η) as follows:

$$(\xi, \eta) = A_{(x_0, y_0, \theta_0)}^{-1}(x, y).$$

This provides the relation btwn $\Psi_{(0,0,0)}$ and $\Psi_{(x_0, y_0, \theta_0)}$:

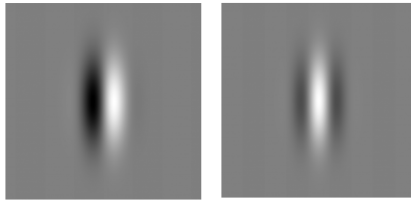
$$\Psi_{(0,0,0)}(x, y) = \frac{1}{2 \pi \sigma^2} \exp\left(-\frac{x^2 + y^2}{\sigma^2}\right) \exp\left(i \frac{y}{\sigma}\right)$$

$$\Psi_{(x_0, y_0, \theta_0)}(x, y) = \Psi_{(0,0,0)}(A_{(x_0, y_0, \theta_0)}^{-1}(x, y))$$



Modeling simple cell receptive profiles

odd (sin)
component even (cos)
component



Gabor functions

$$\Psi_{(x_0, y_0, \theta_0)}(x, y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{\xi^2 + \eta^2}{\sigma^2}\right) \exp\left(i\frac{\eta}{\sigma}\right)$$

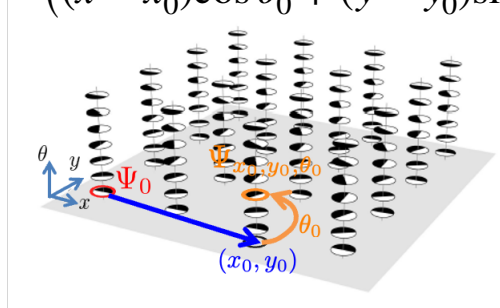
Why Gabor functions are preferred as RP models in neurogeometric approach?

- Gabor functions are generated from a reference Gabor function (Gabor *mother* function):

$$\Psi_{(x_0, y_0, \theta_0)}(x, y) = \Psi_{(0, 0, 0)}(A_{(x_0, y_0, \theta_0)}^{-1}(x, y))$$

$$\begin{pmatrix} \xi \\ \eta \end{pmatrix} = R_{\theta_0}^{-1} \begin{pmatrix} x - x_0 \\ y - y_0 \end{pmatrix}$$

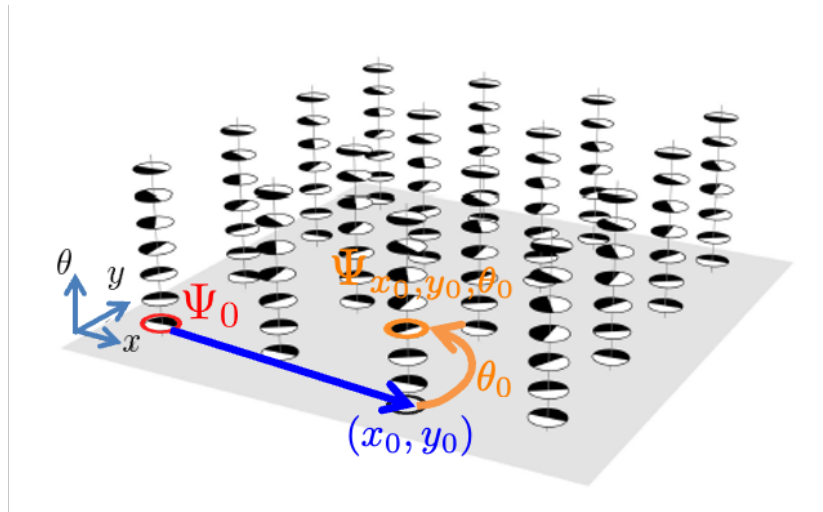
$$A_{(x_0, y_0, \theta_0)}^{-1}(x, y) = (\xi, \eta) = ((x - x_0)\cos\theta_0 + (y - y_0)\sin\theta_0, -(x - x_0)\sin\theta_0 + (y - y_0)\cos\theta_0)$$



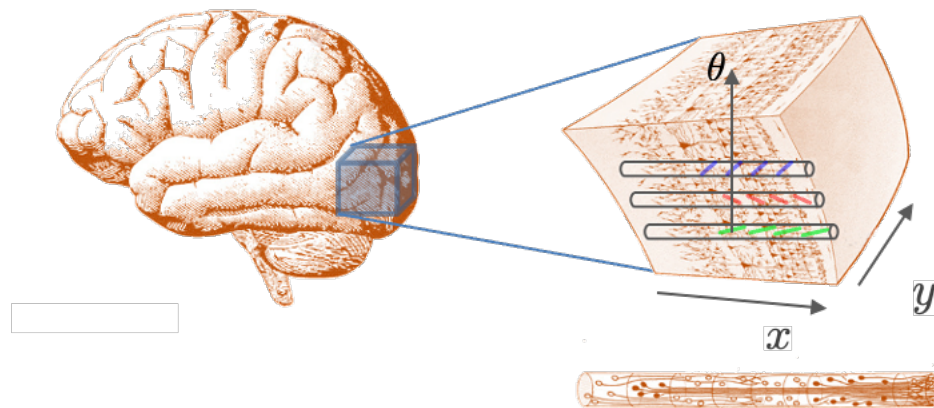
Set of receptive profiles

- They are minimizers of the uncertainty arising from simultaneous detection of position and orientation.

Functional architecture of the primary visual cortex



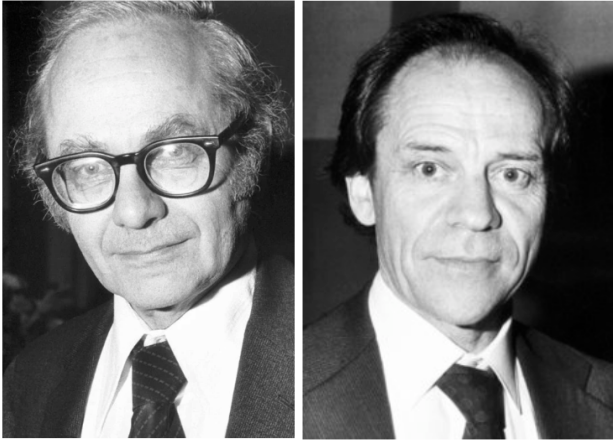
Set of receptive profiles



V1 columnar structure

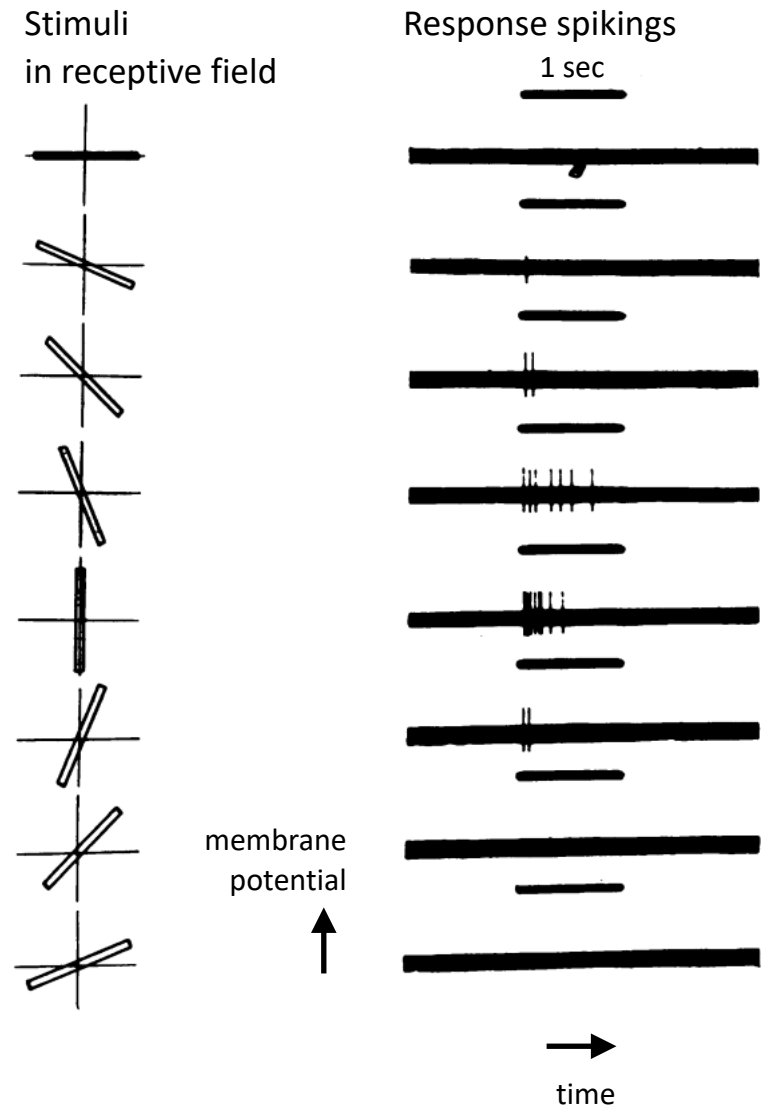
Functional architecture of the primary visual cortex

David H. Hubel & Torsten N. Wiesel



nobelprize.org

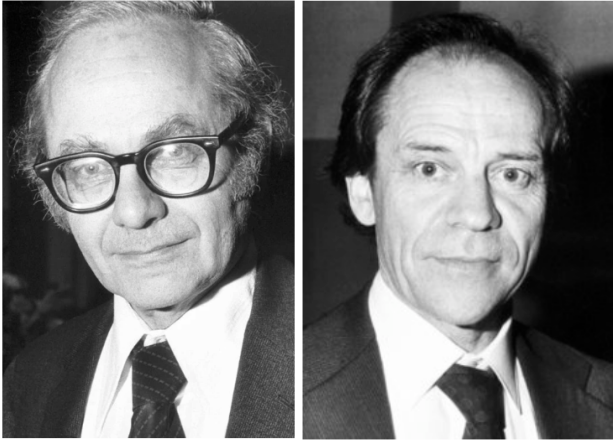
Orientation selectivity of simple cells



Hubel & Wiesel, J. Physiology, 1959

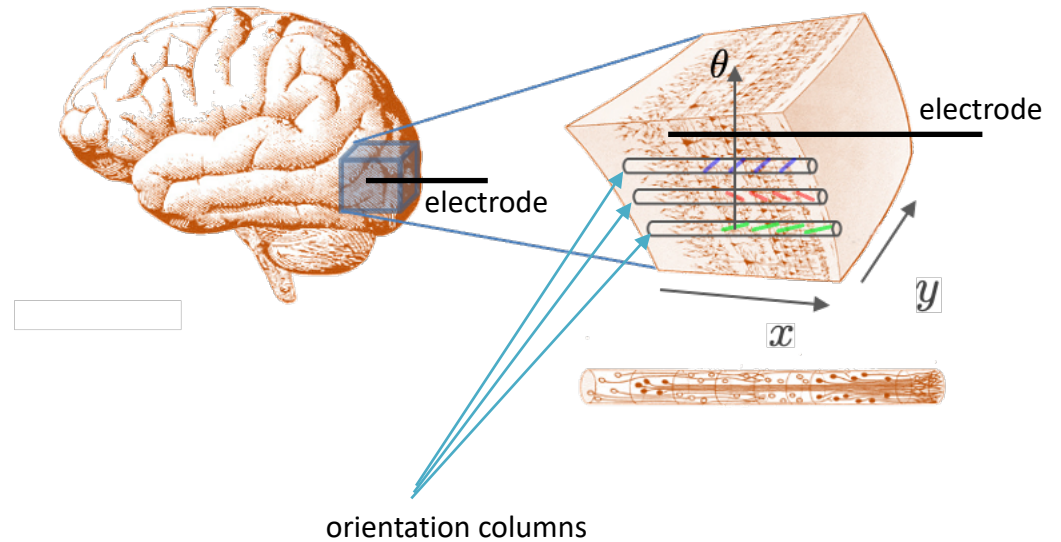
Functional architecture of the primary visual cortex

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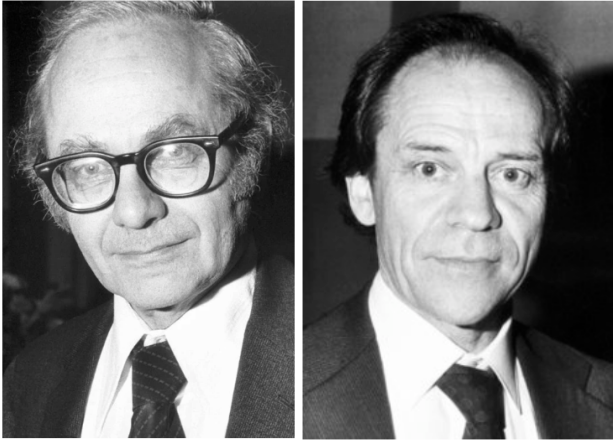
[nobelprize.org](https://www.nobelprize.org)

Columnar organization of simple cells



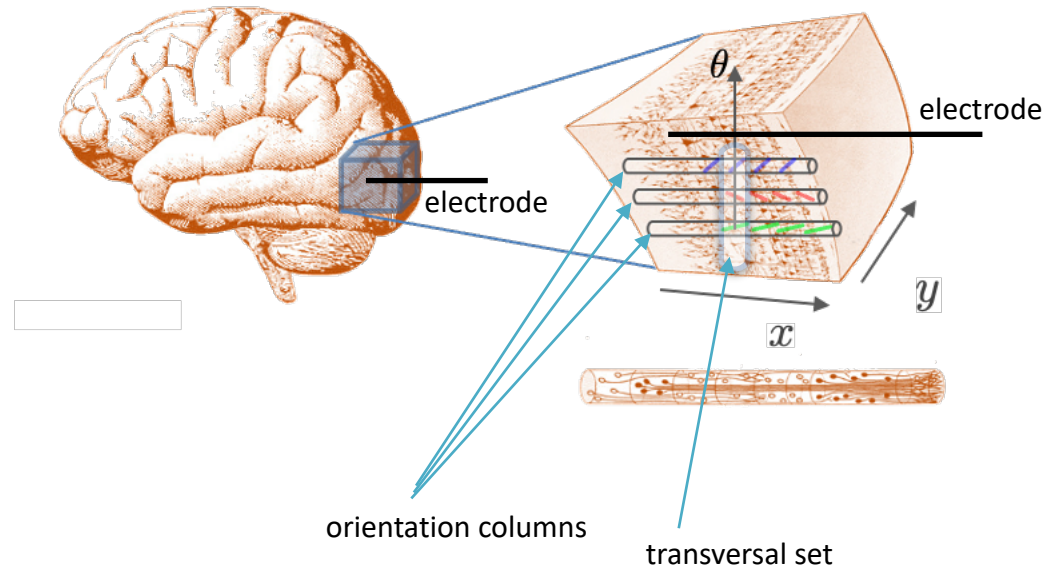
Functional architecture of the primary visual cortex

David H. Hubel & Torsten N. Wiesel



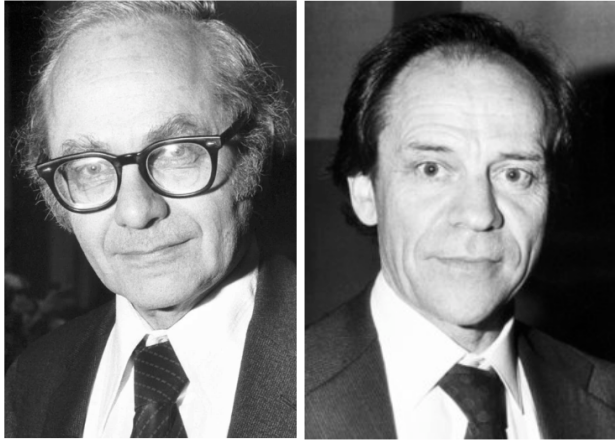
[nobelprize.org](https://www.nobelprize.org)

Columnar organization of simple cells

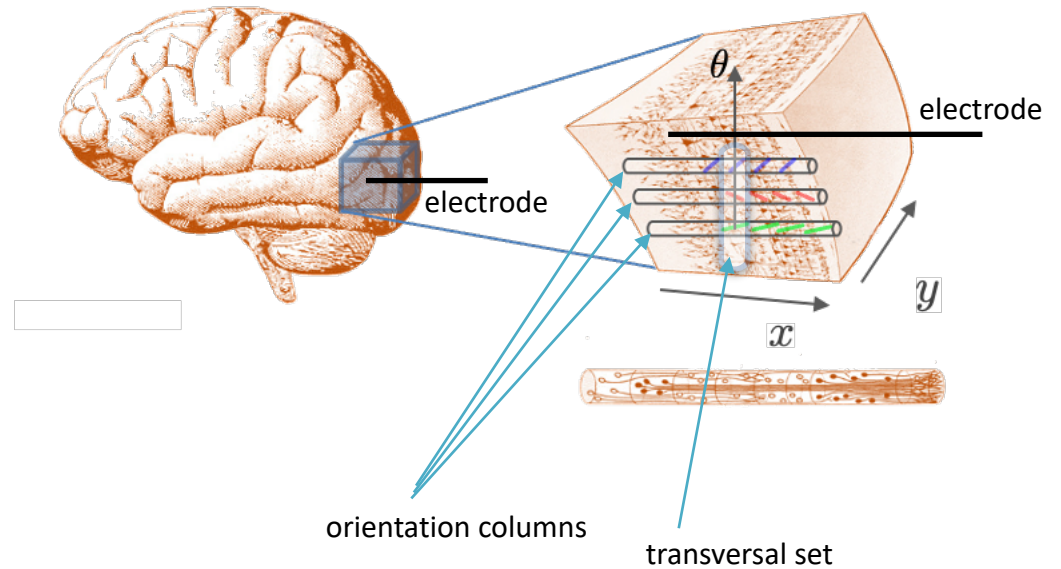


Functional architecture of the primary visual cortex

David H. Hubel & Torsten N. Wiesel



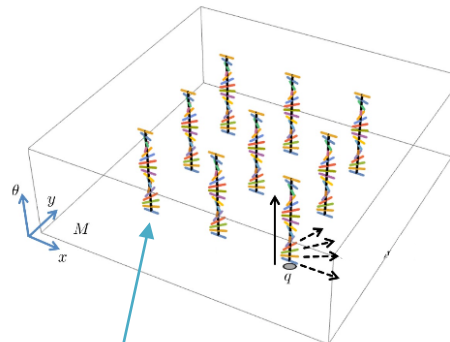
nobelprize.org



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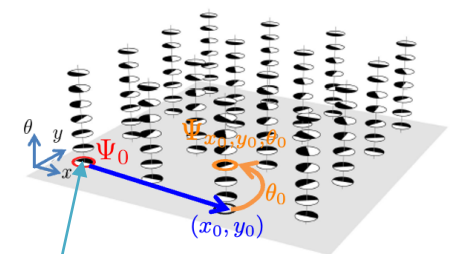
- Transversal sets are distributed in a periodic (spatial) fashion on the cortical surface.
- This is similar to the set of receptive profiles.
- This is the key point linking the columnar functional architecture of the primary visual cortex to the set of receptive profiles.

neurogeometric setting



orientation fiber

set of receptive profiles



receptive profiles
on an orientation fiber

Citti-Sarti-Petitot model of the functional architecture

Model assumptions:

- No motion. 2D grayscale images are assumed as visual input.
 - Restricted to a single instant.
 - All kinds of movements (head, eye) are excluded, not only the movements in a visual scene.
 - Therefore, no dynamics are taken into account. It is a static model.

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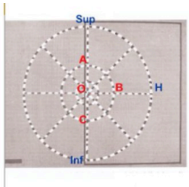
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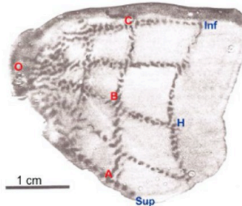
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retinal surface



cortical surface



retino-cortical
mapping

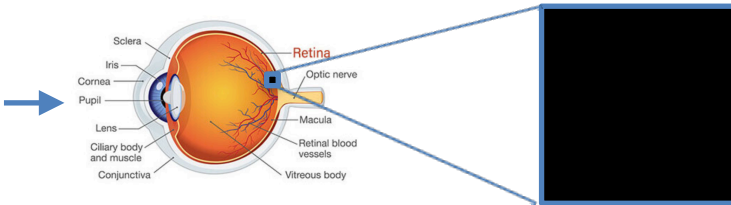
retinal surface $\neq$ cortical surface

image plane



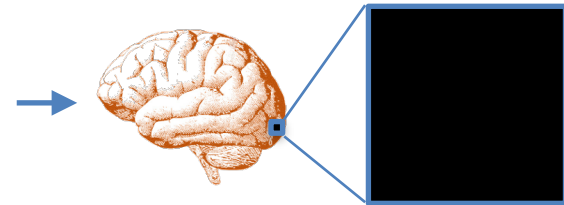
$$P \subset \mathbb{R}^2$$

retinal plane



$$R \subset \mathbb{R}^2$$

V1 plane

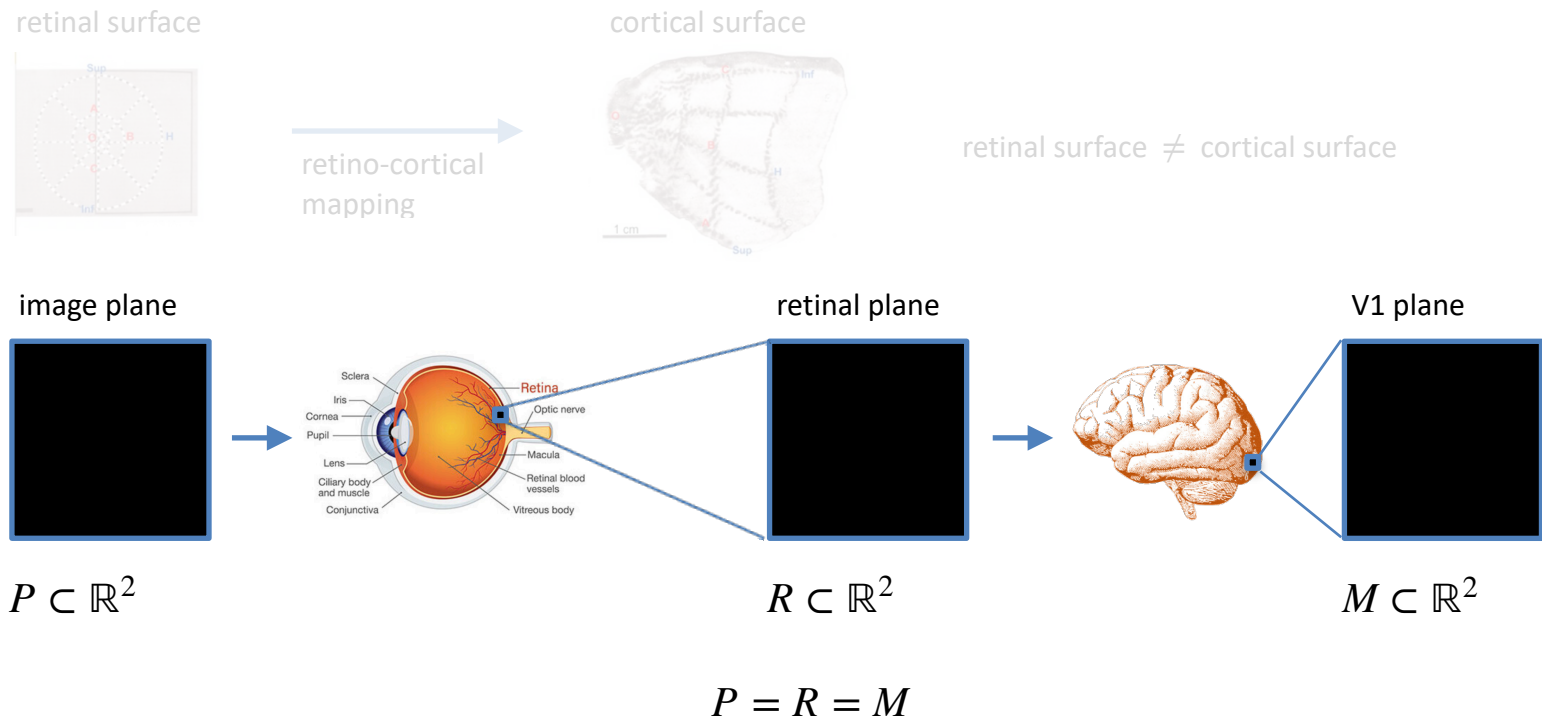


$$M \subset \mathbb{R}^2$$

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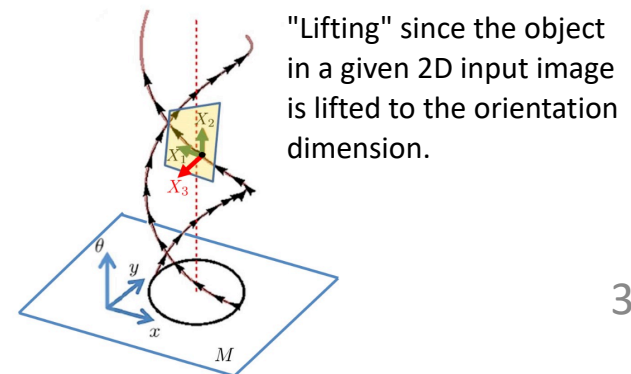
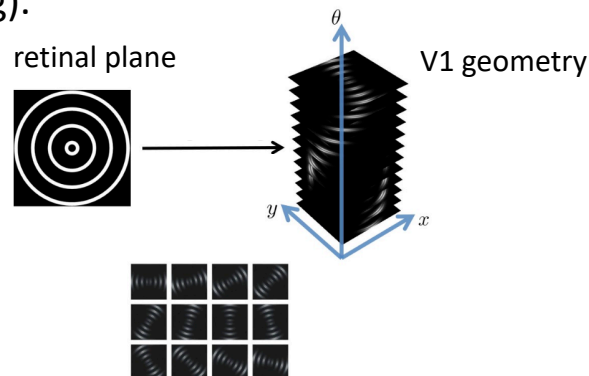
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Focuses only on orientation-selective behavior of the simple cells in V1!

Selectivity to other visual features (scale, spatial frequency etc.) is not taken into account!
(But extensions to other features exist in the modeling literature)

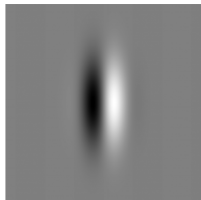
Citti-Sarti-Petitot model of the functional architecture

We restrict ourselves to orientation selectivity!

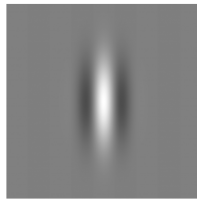
What do we mean by orientation selectivity?

Overlap of orientation of a simple cell receptive profile with the orientation of visual input.

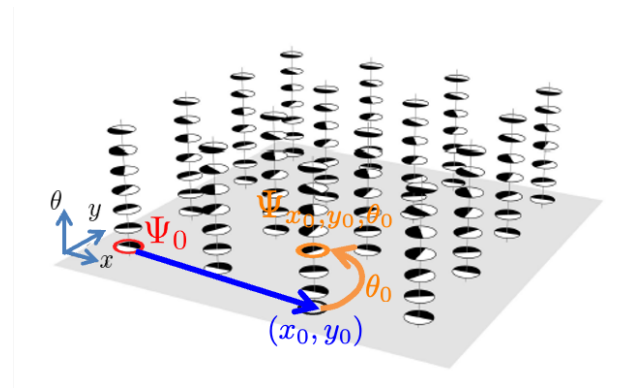
odd (sin)
component



even (cos)
component



Gabor functions



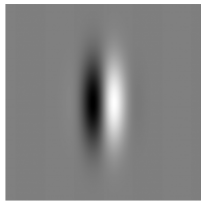
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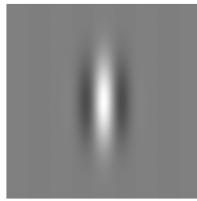
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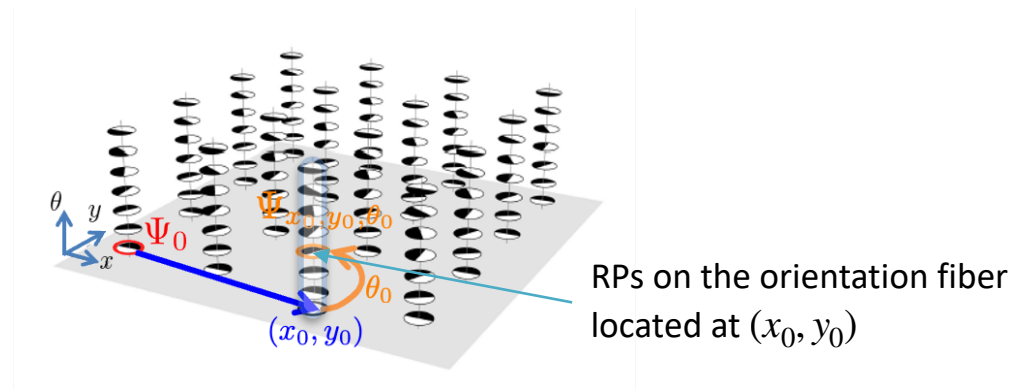
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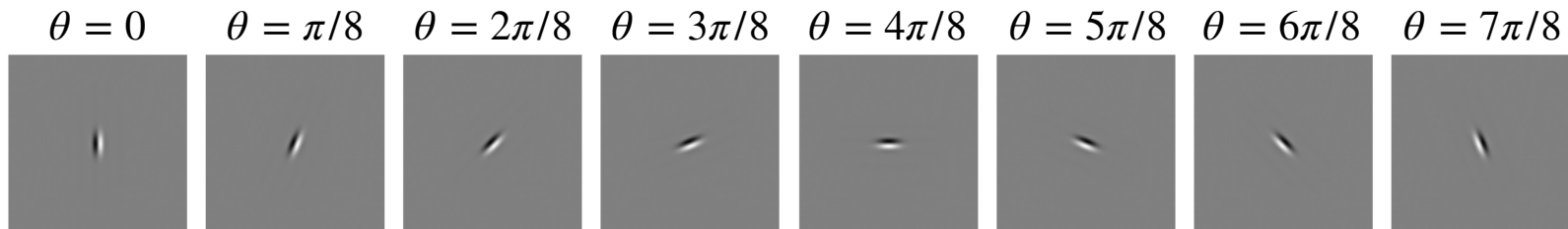


Gabor functions



At (x_0, y_0) , we have **all** the receptive profiles $\in \{\Psi_{(x_0, y_0, \theta_0)} : \theta_0 \in [0, \pi), (x_0, y_0) \in \mathbb{R}^2\}$.

They look like

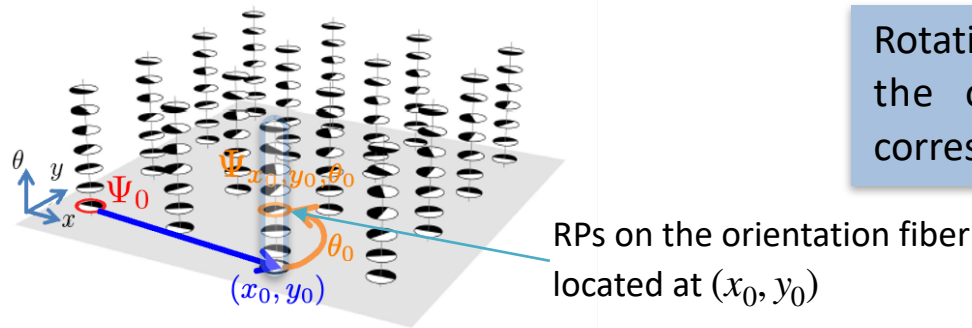


Citti-Sarti-Petitot model of the functional architecture

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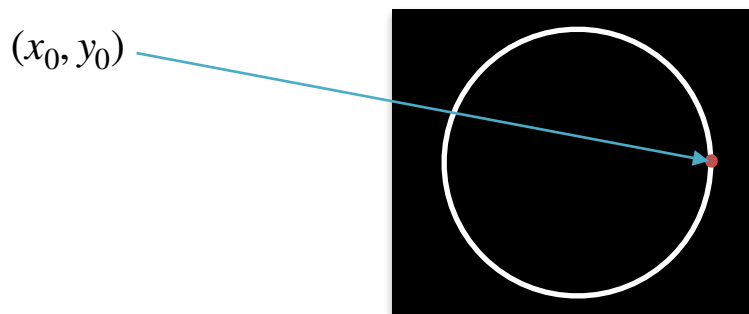
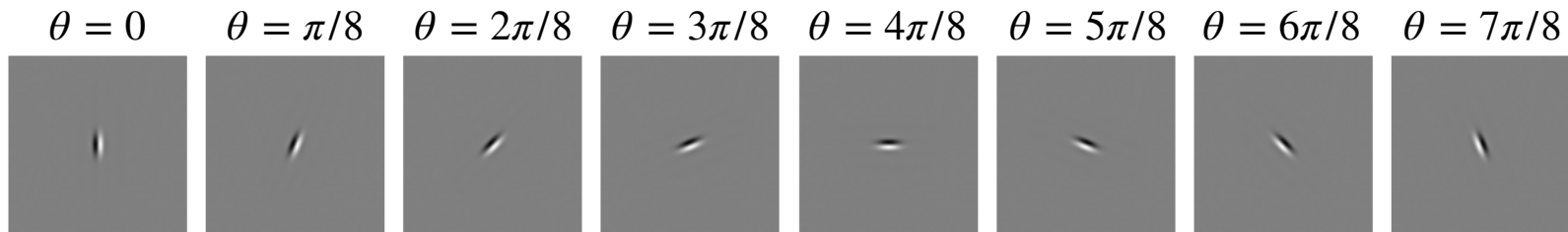
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How is that?

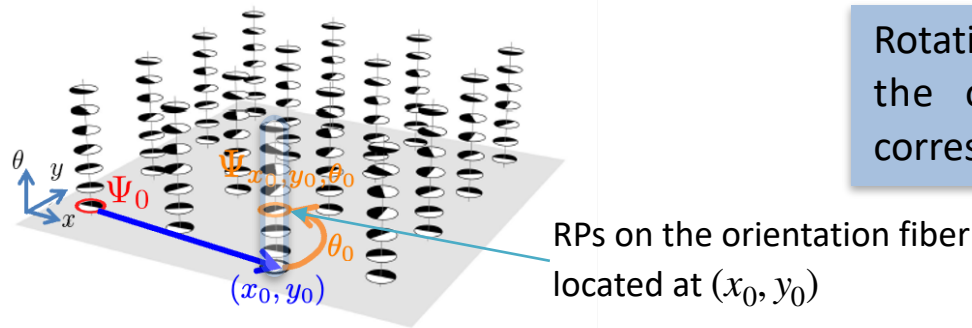


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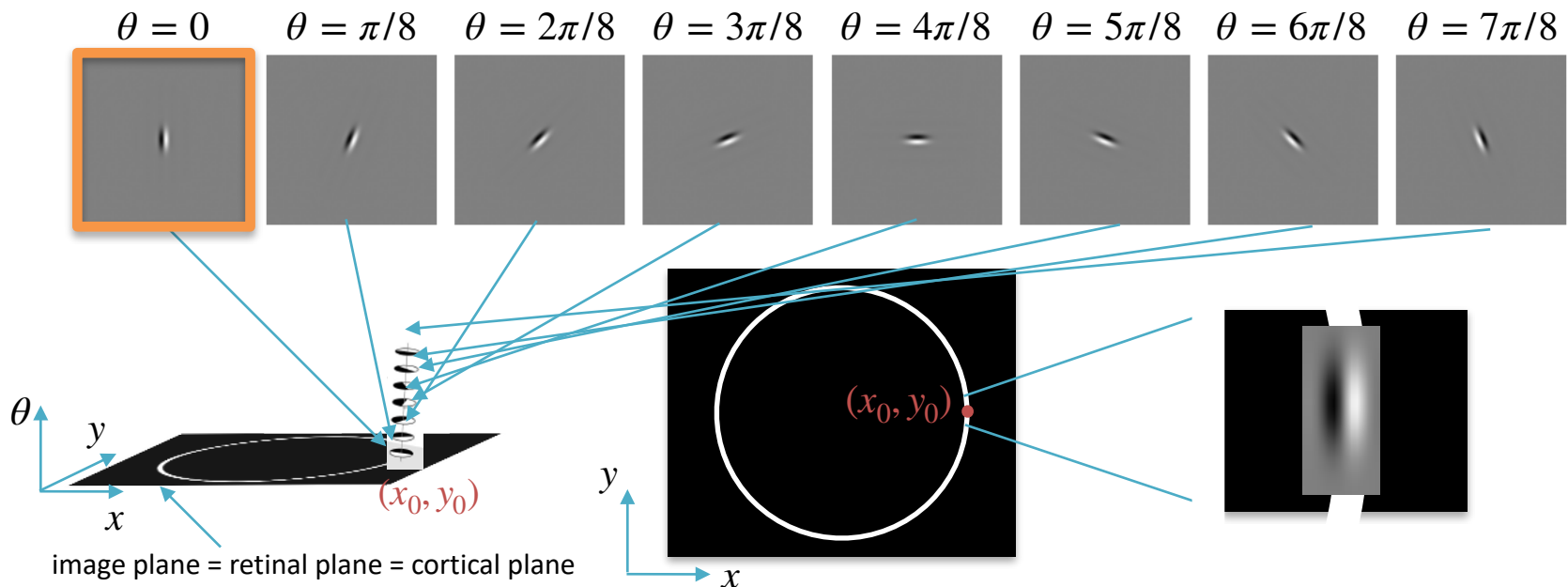
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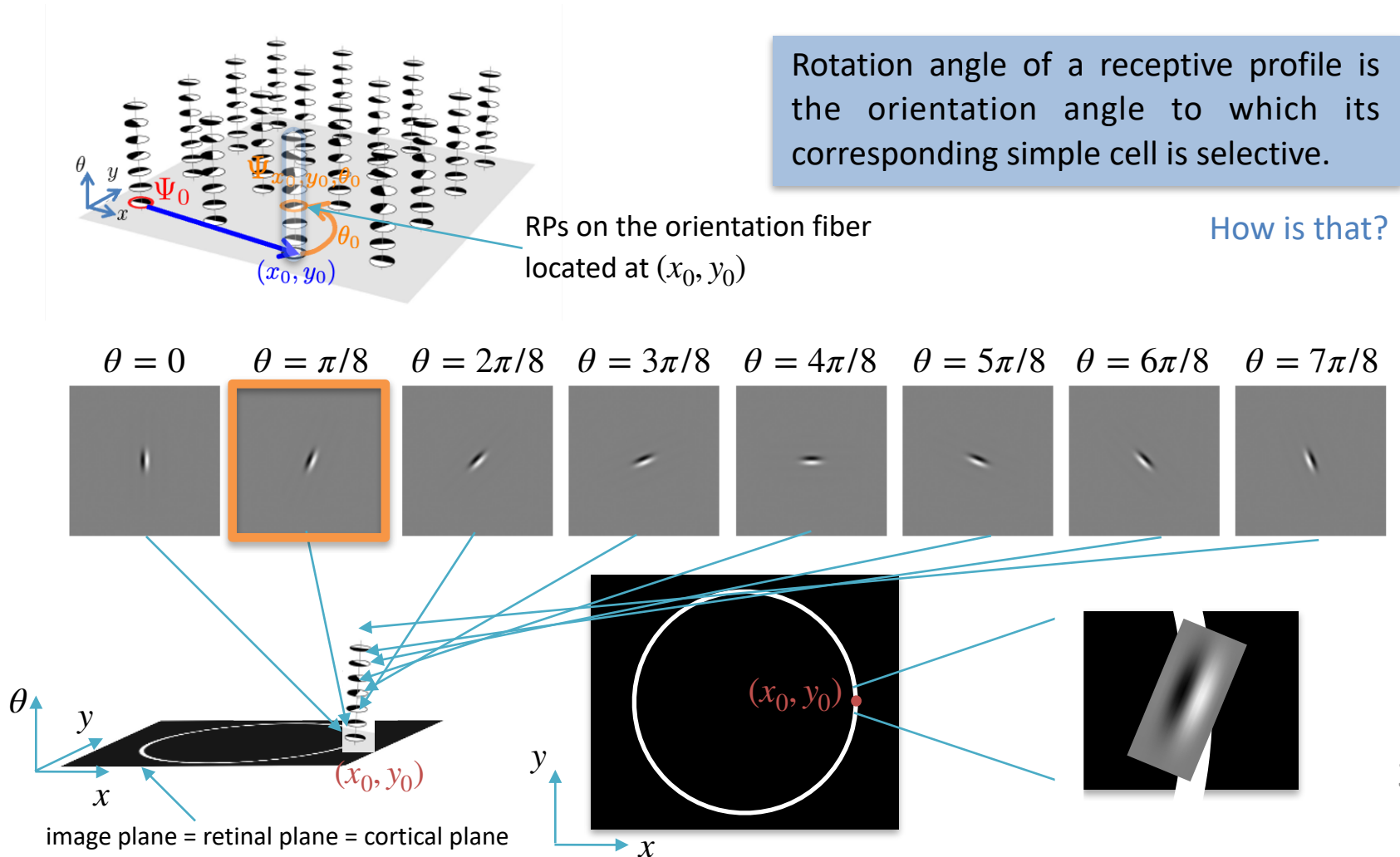


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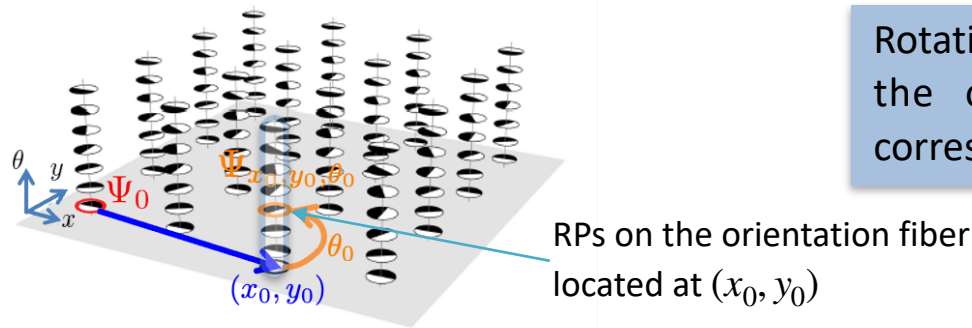


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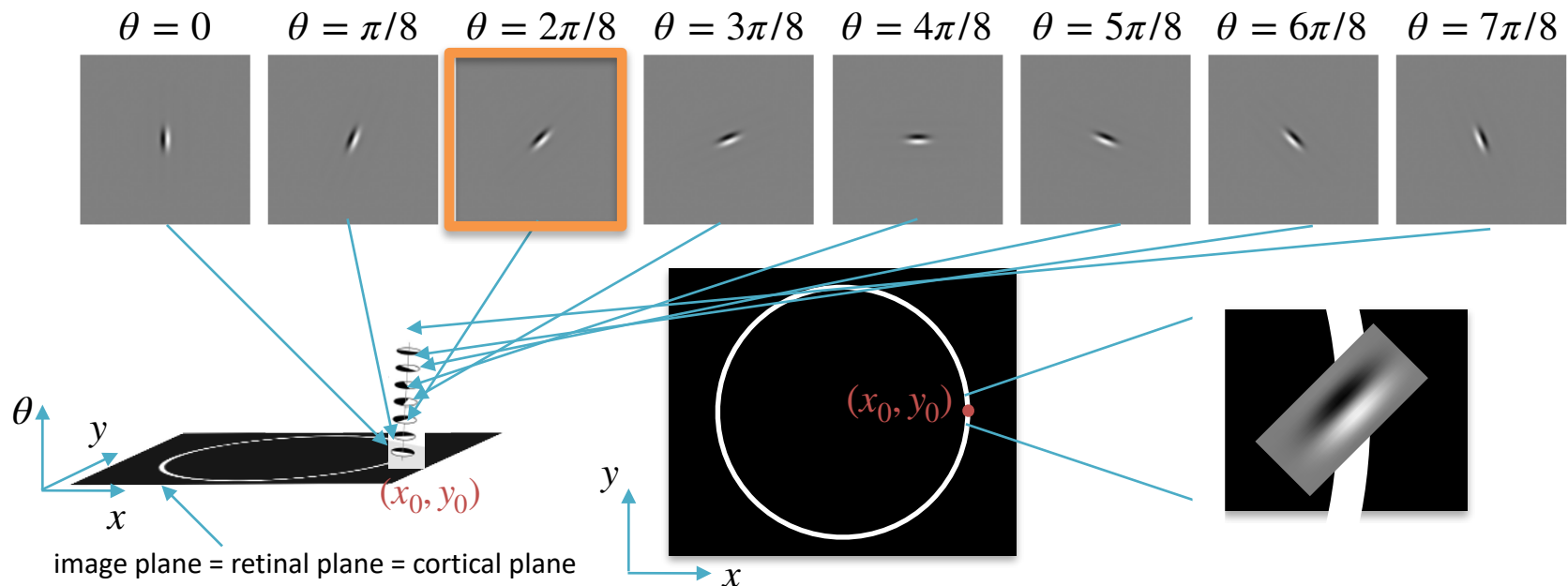
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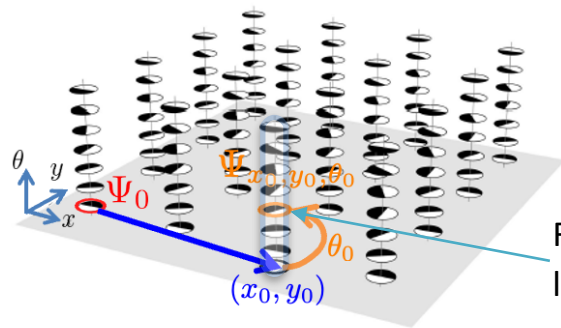


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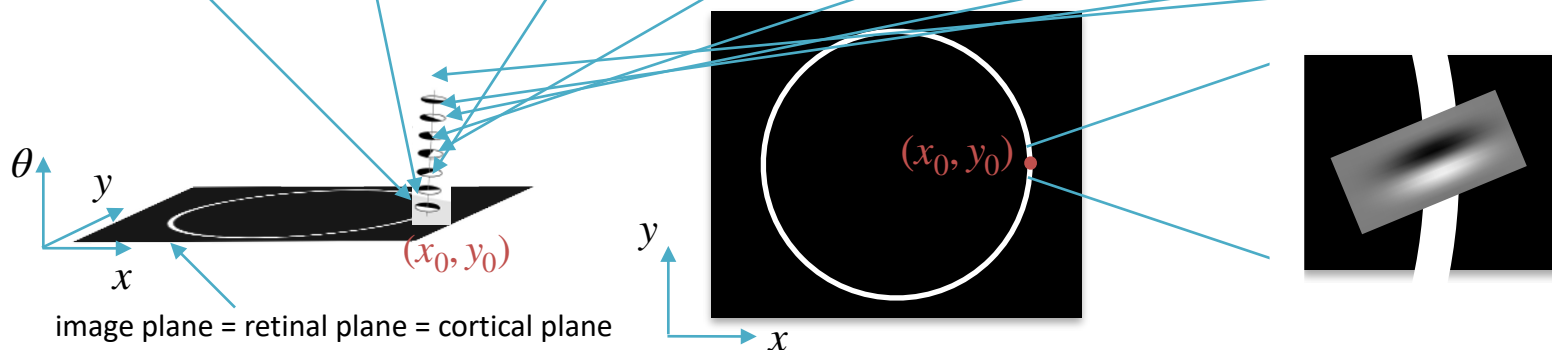
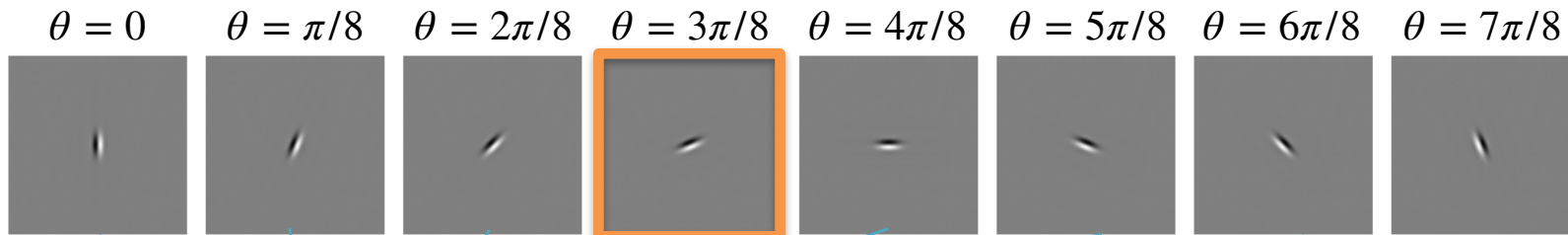
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RPs on the orientation fiber located at (x_0, y_0)

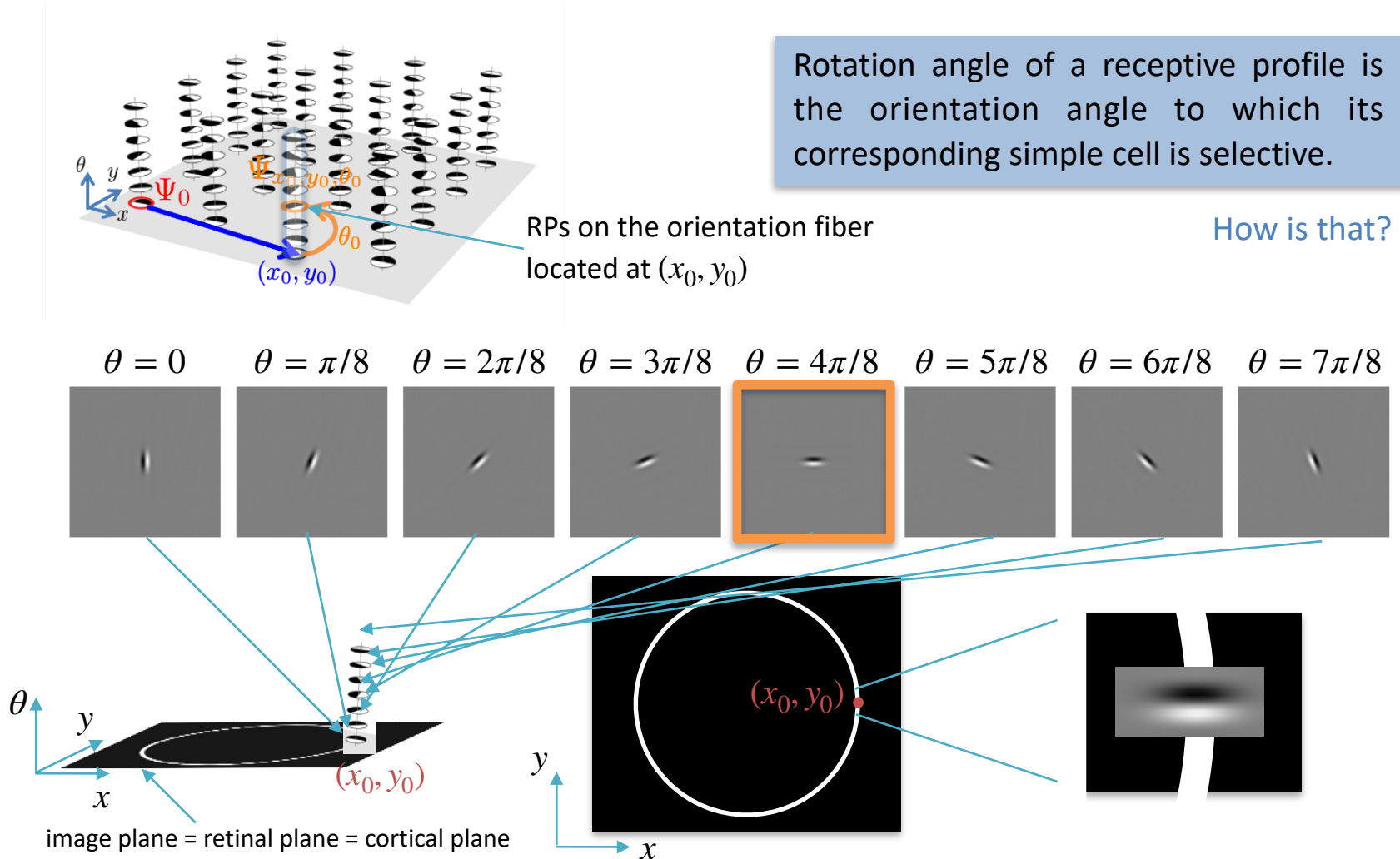


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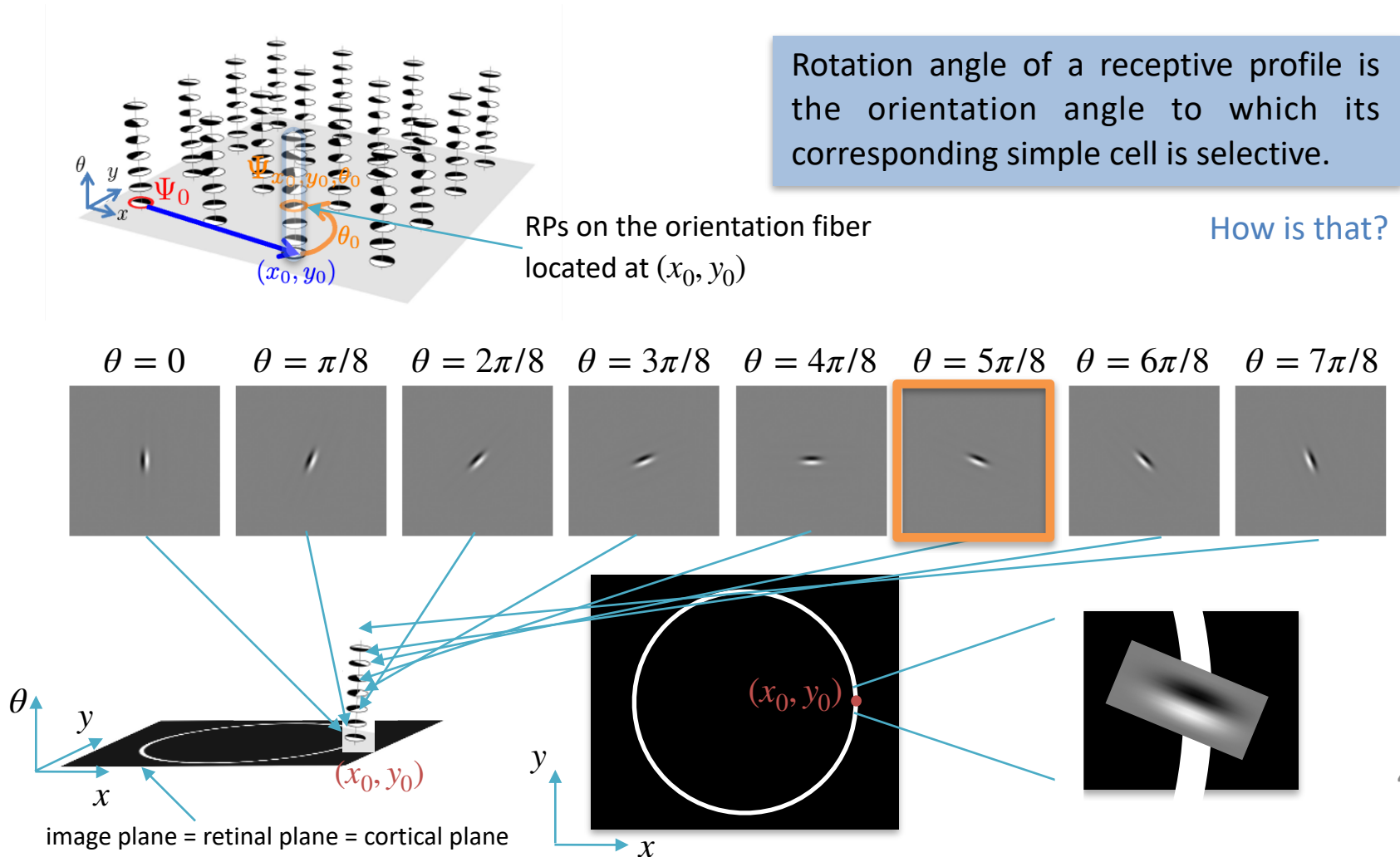


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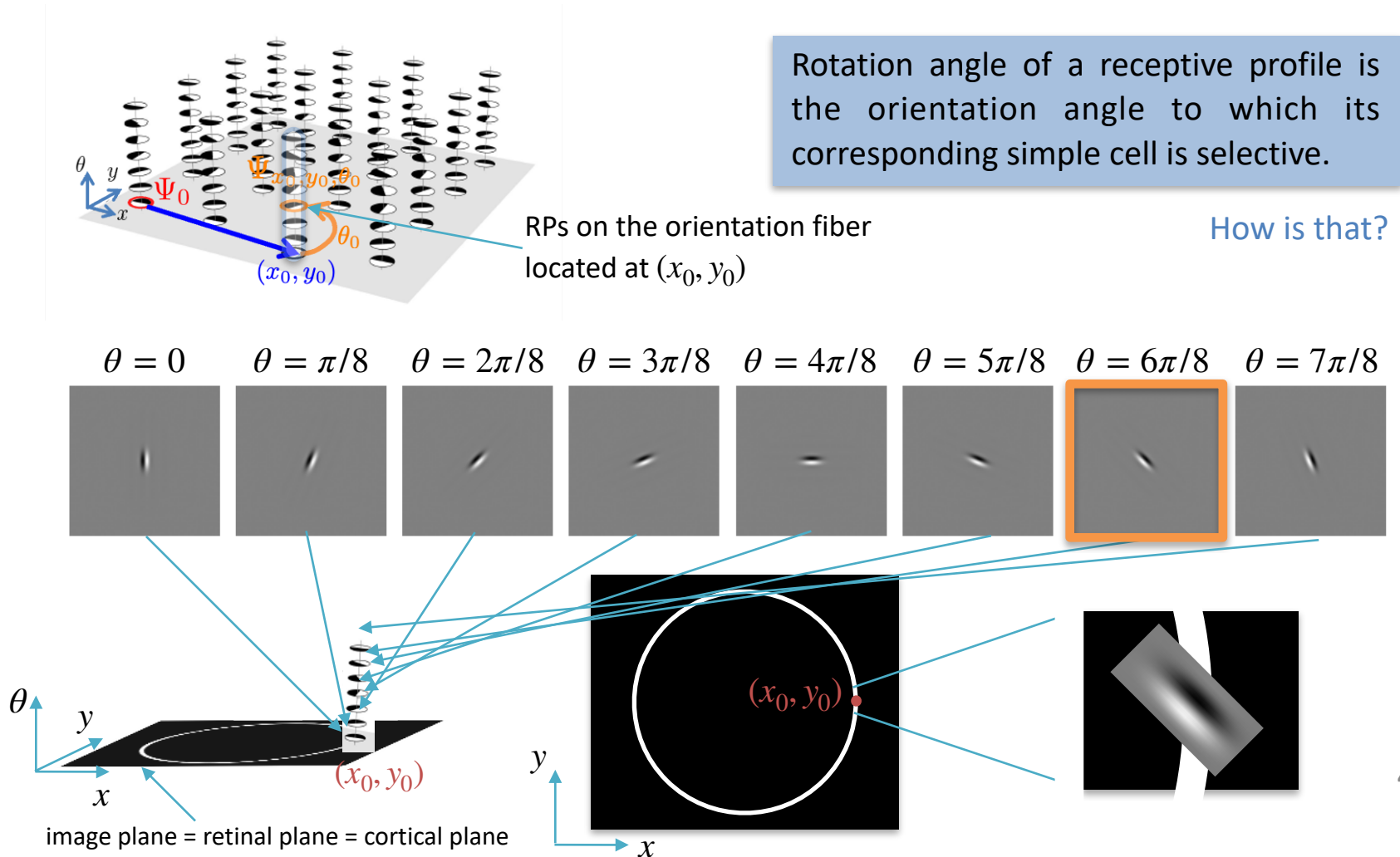


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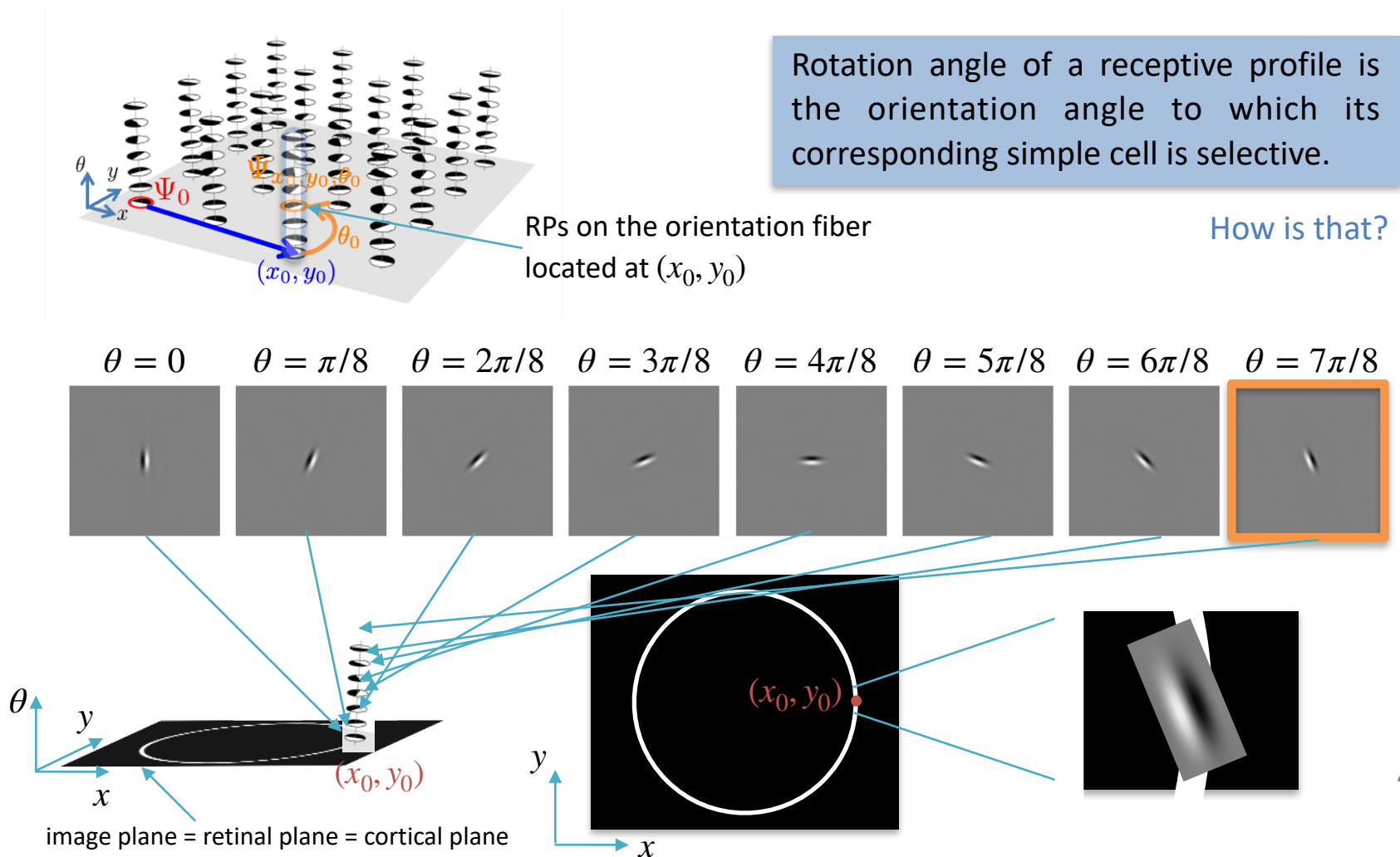


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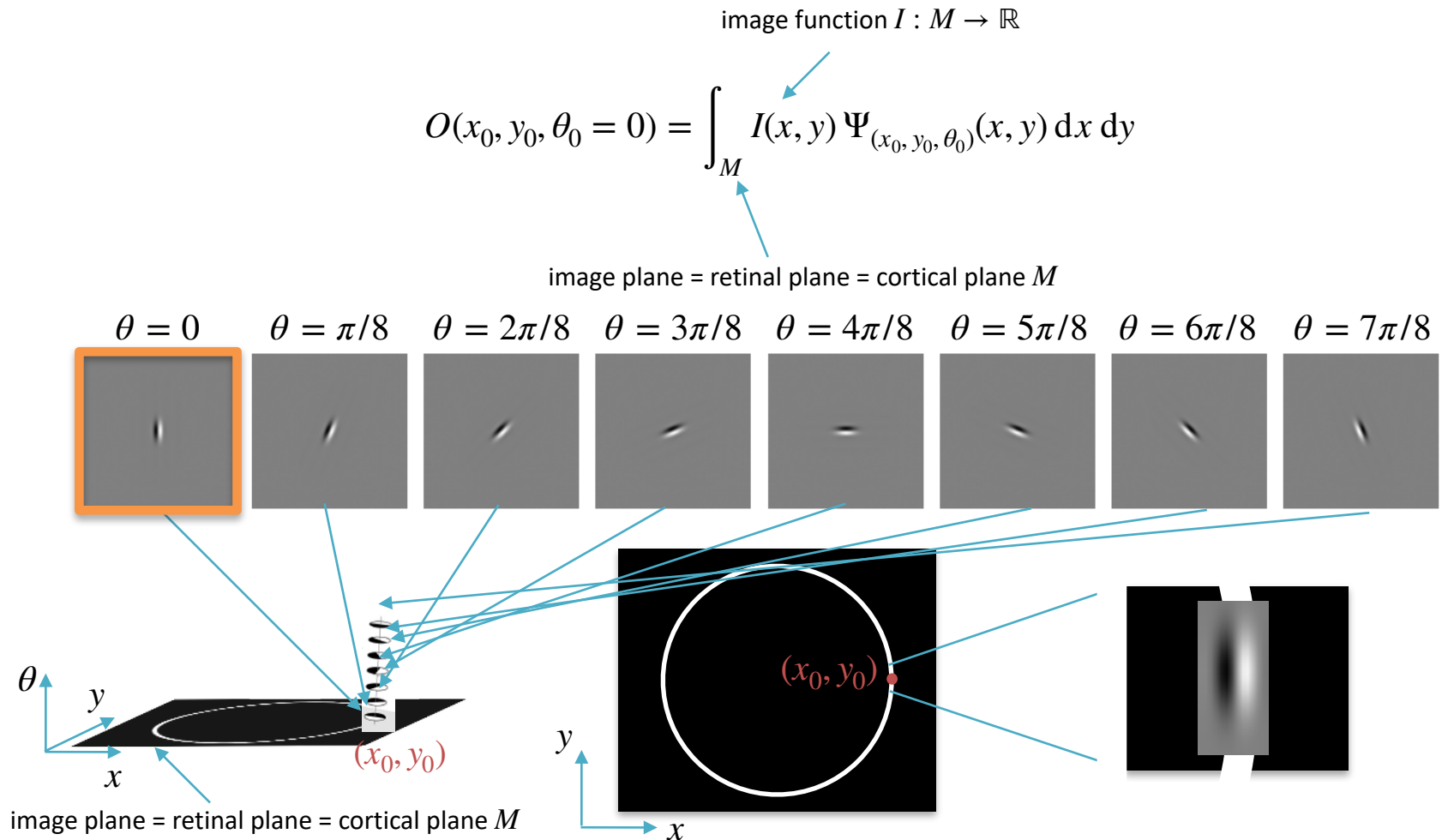


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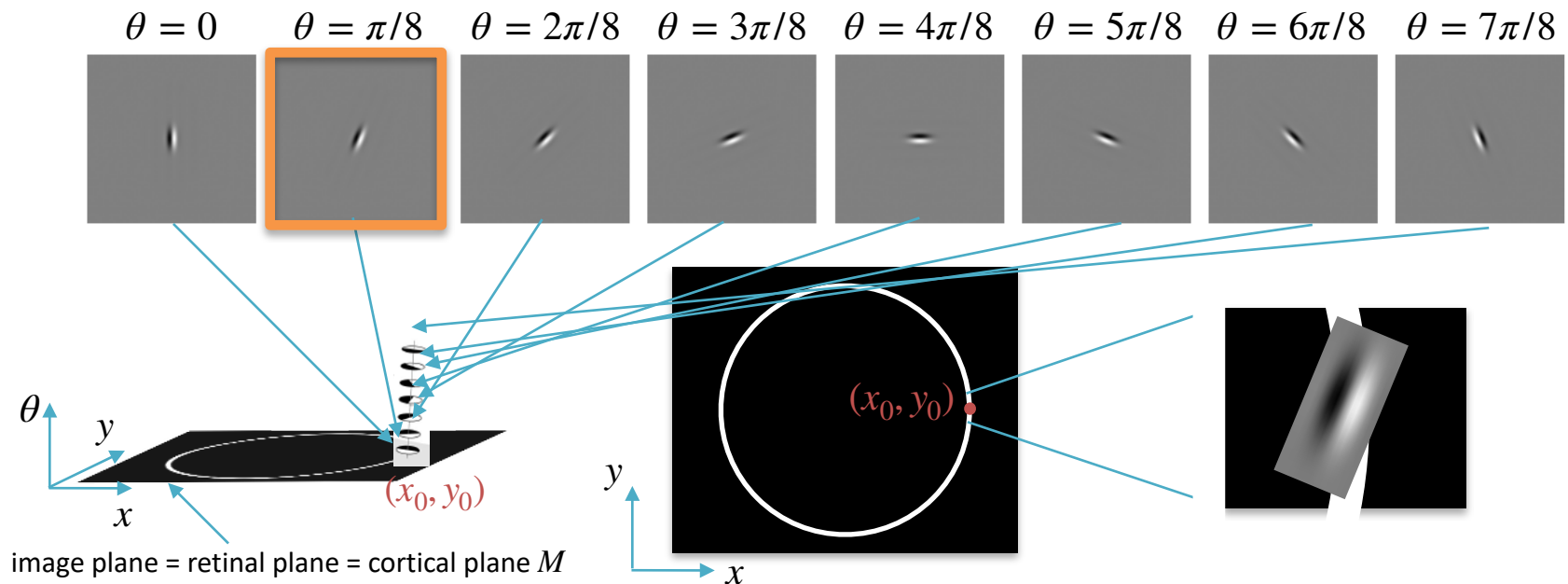
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$$O(x_0, y_0, \theta_0 = \pi/8) = \int_M I(x, y) \Psi_{(x_0, y_0, \theta_0)}(x, y) dx dy$$



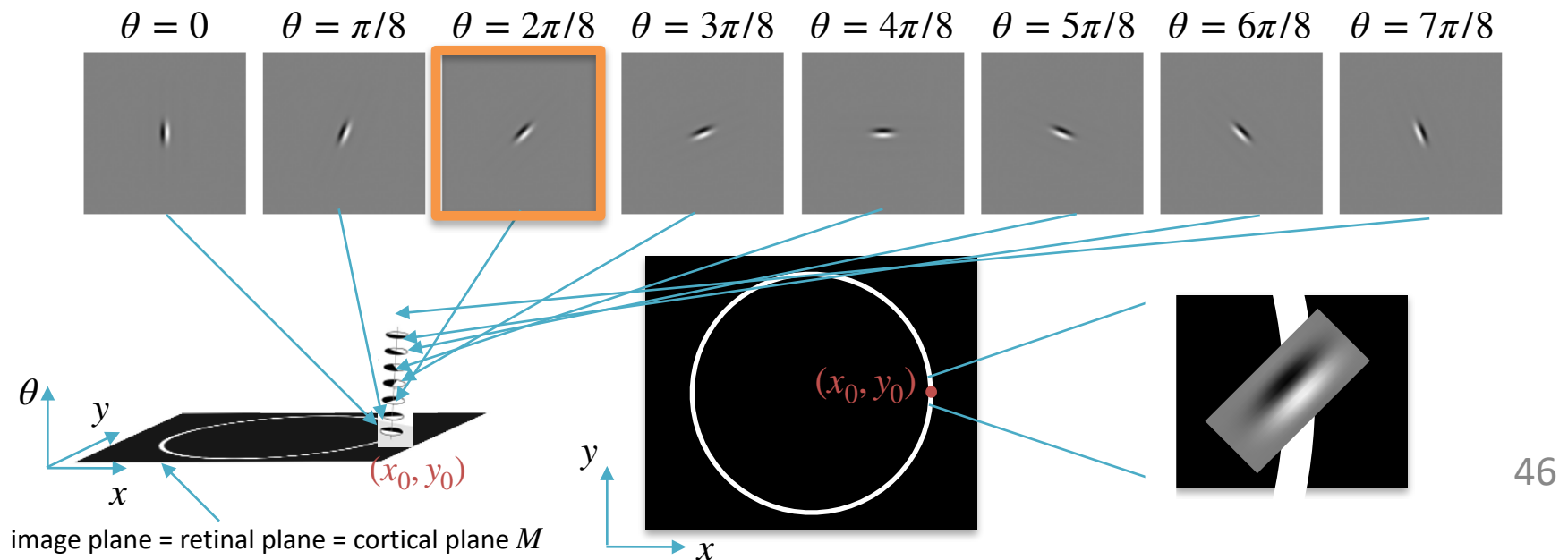
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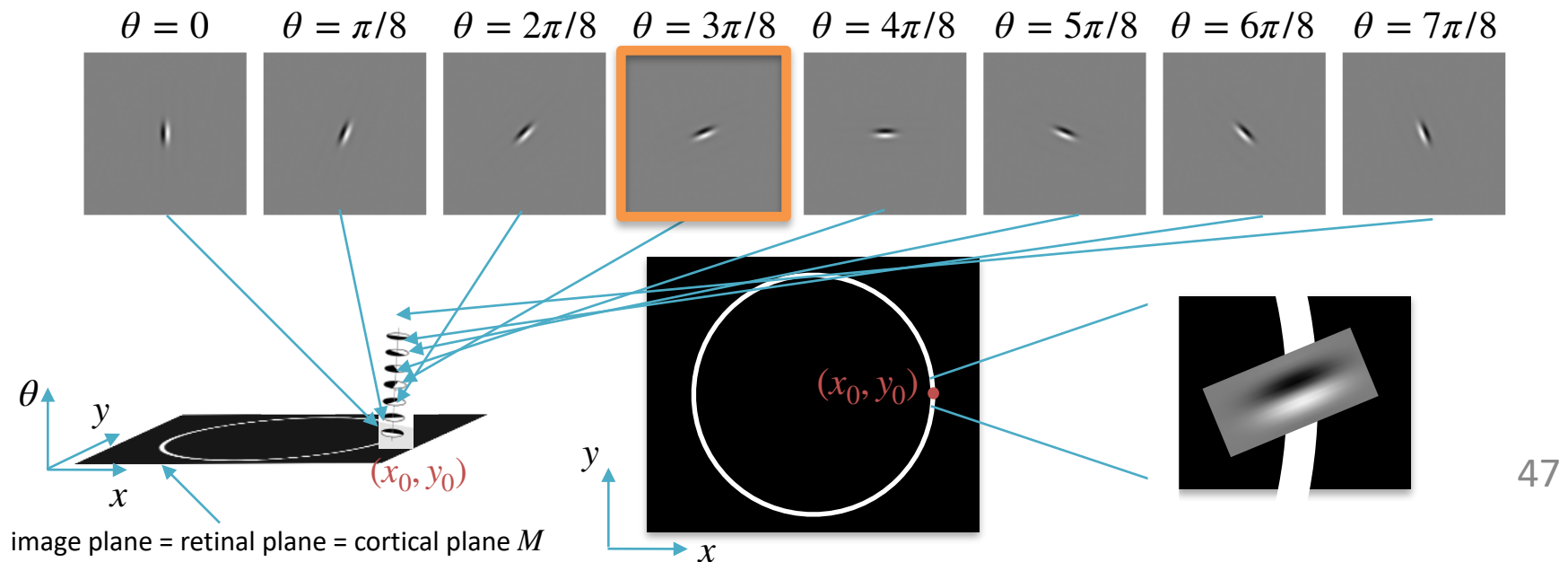
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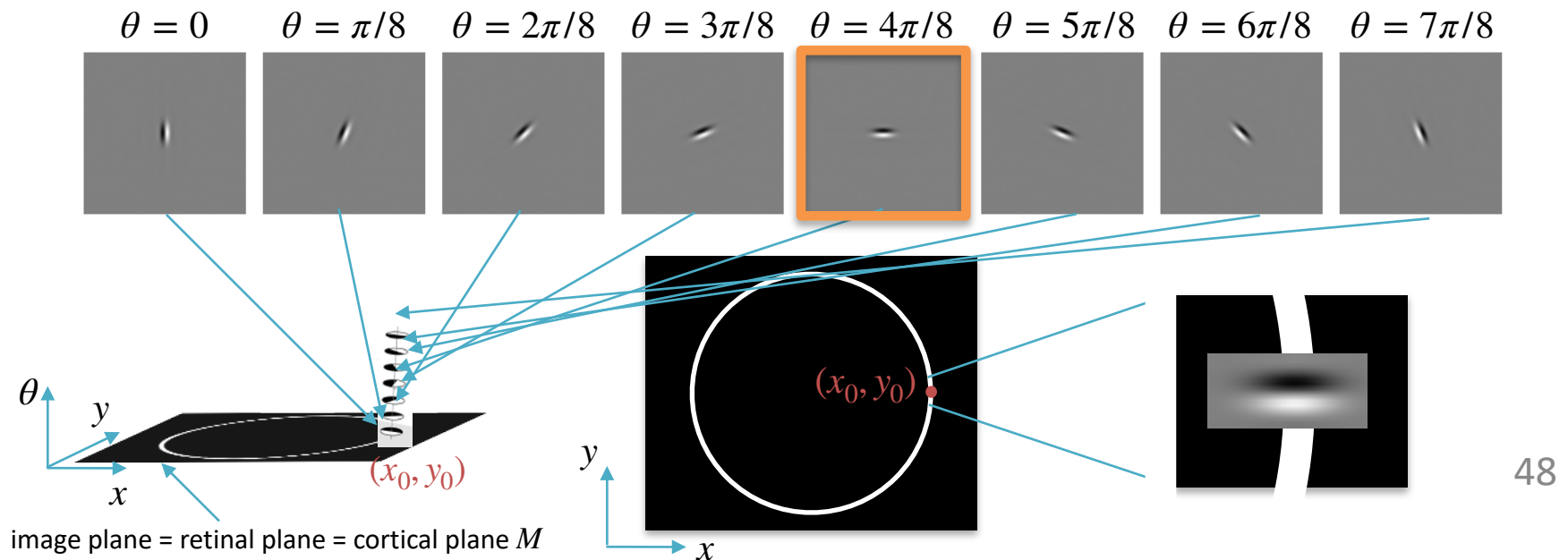
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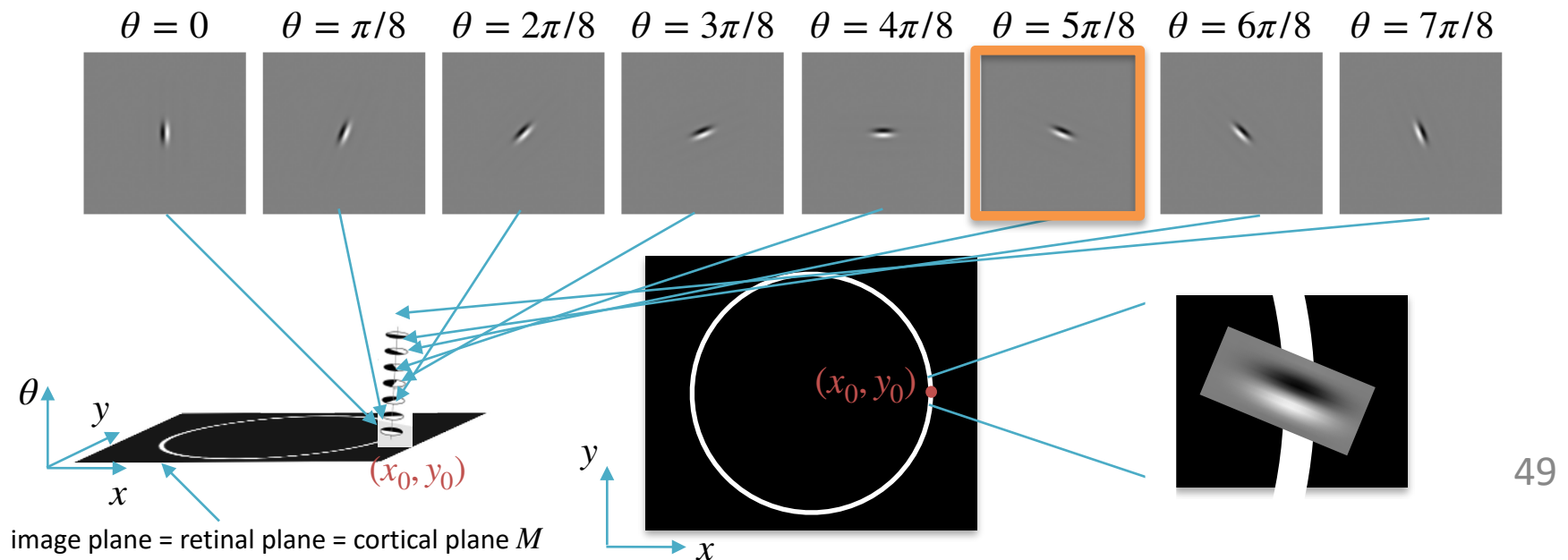
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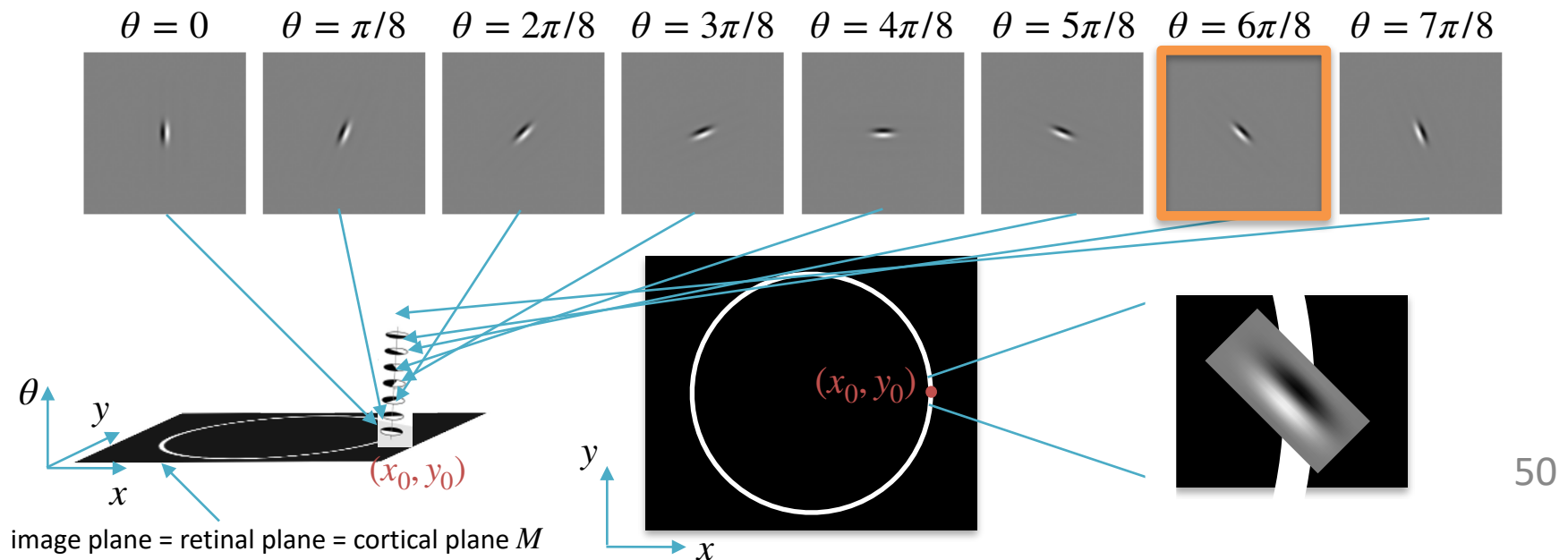
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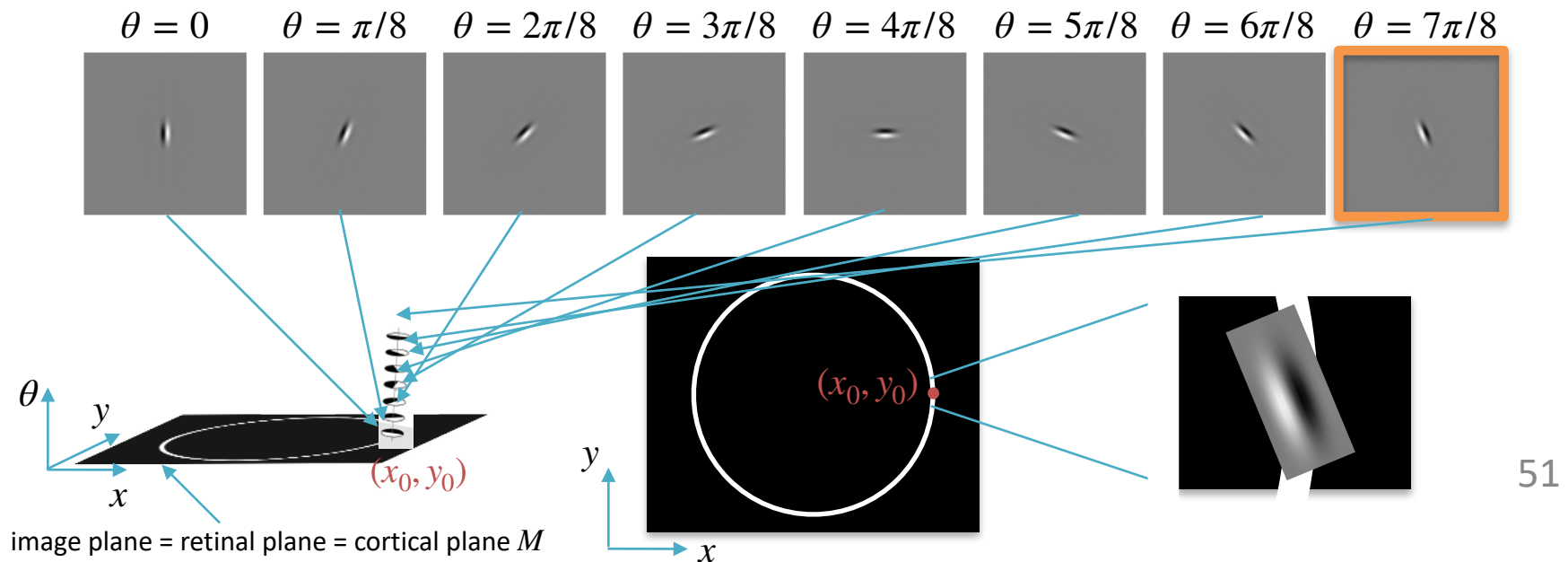
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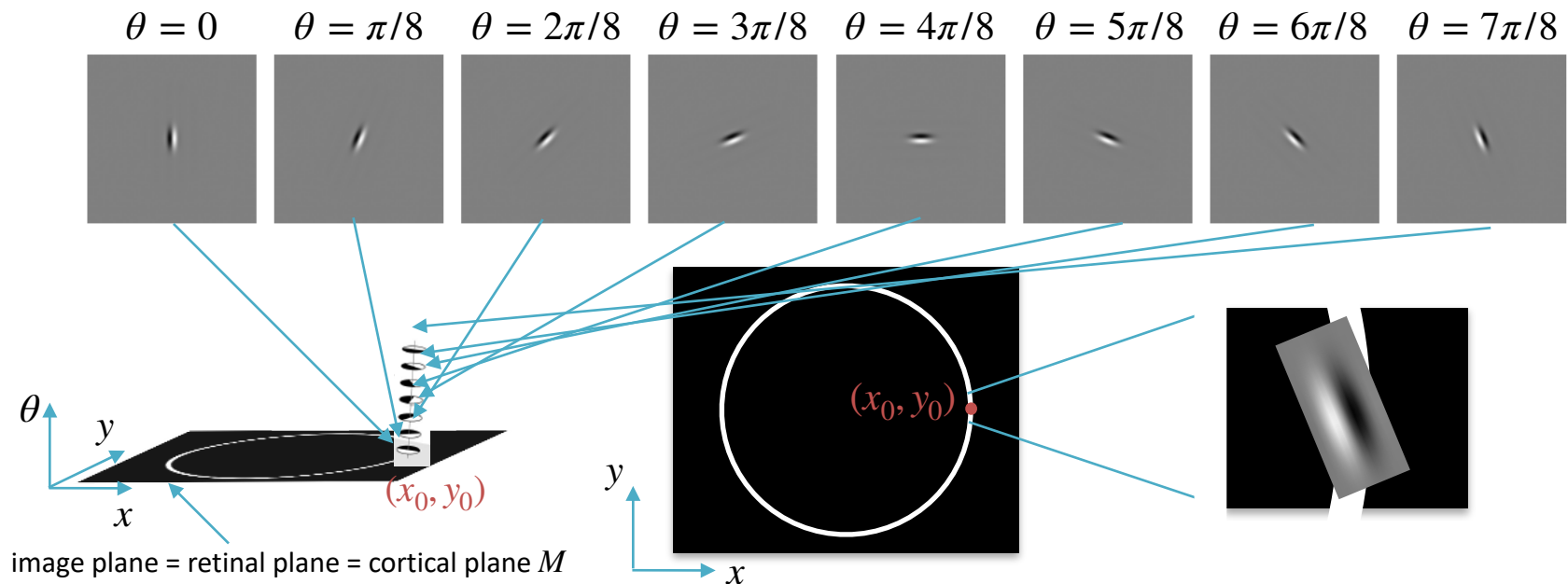
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Generalize to all $(x_0, y_0) \in M$ and $\theta_0 \in [0, \pi)$!



Citti-Sarti-Petitot model of the functional architecture

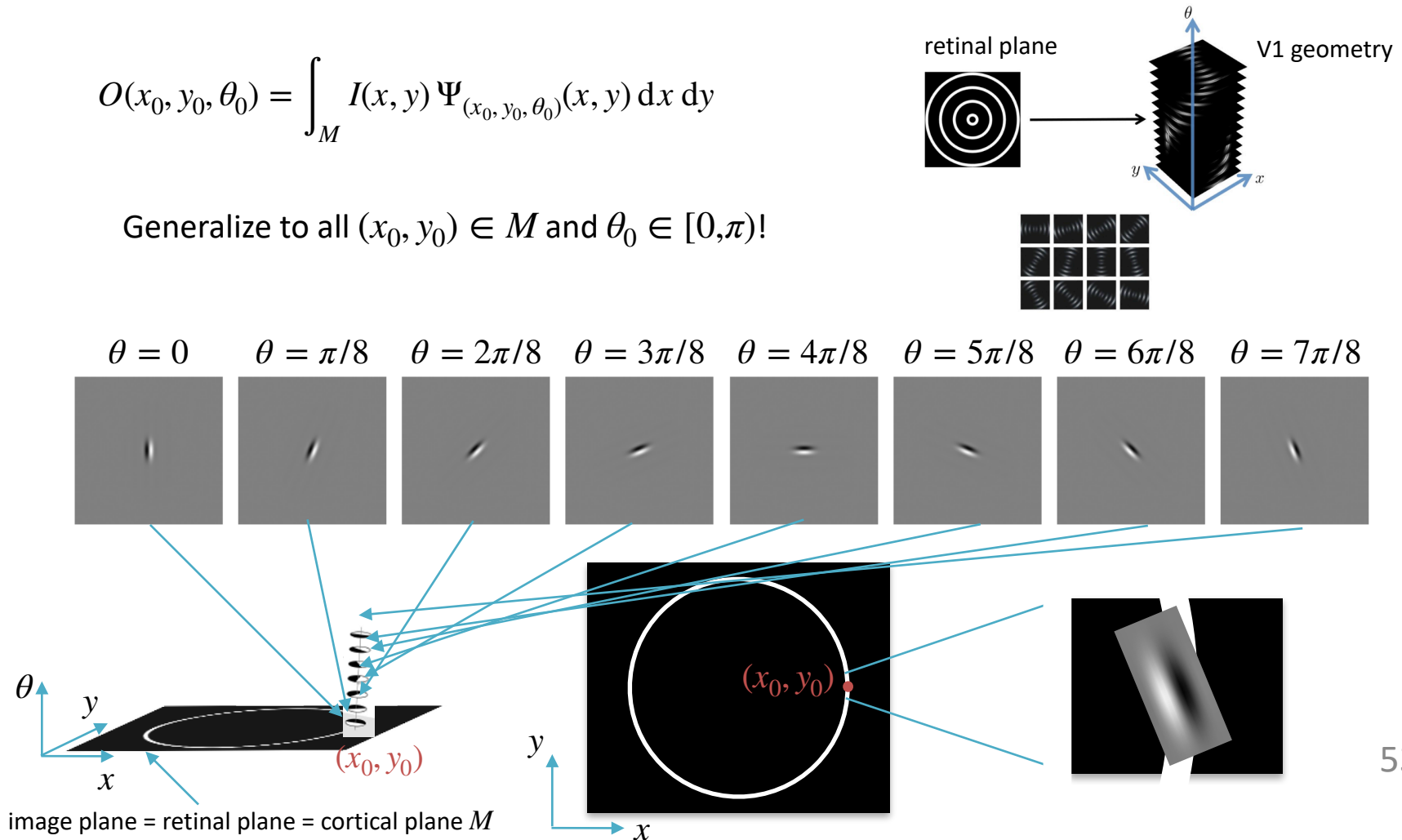
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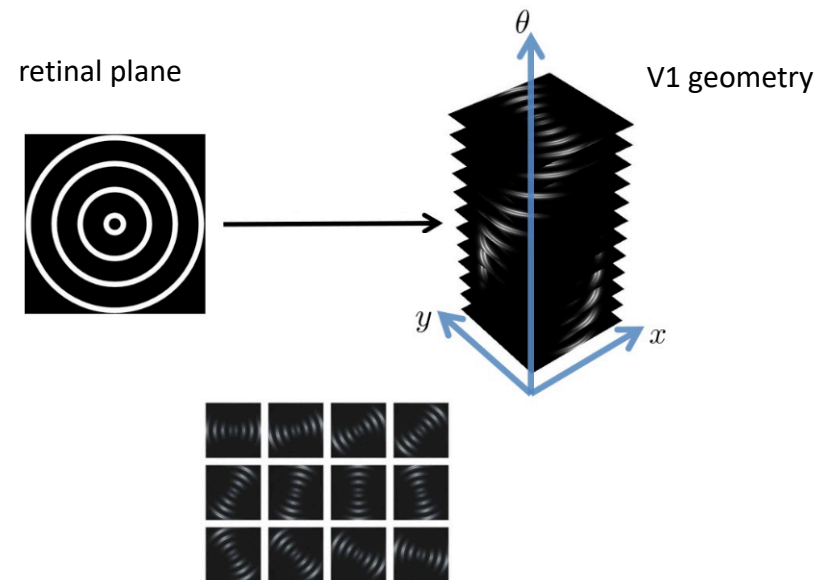
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Output response of the simple cell selective to the position $(x_0, y_0) \in M$ and to the orientation $\theta_0 \in [0, \pi)$



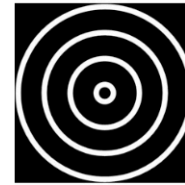
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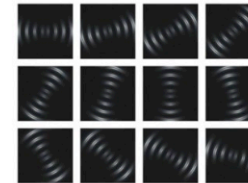
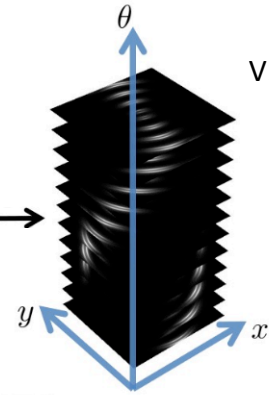


Output response of the simple cell selective to the position $(x_0, y_0) \in M$ and to the orientation $\theta_0 \in [0, \pi)$

retinal plane



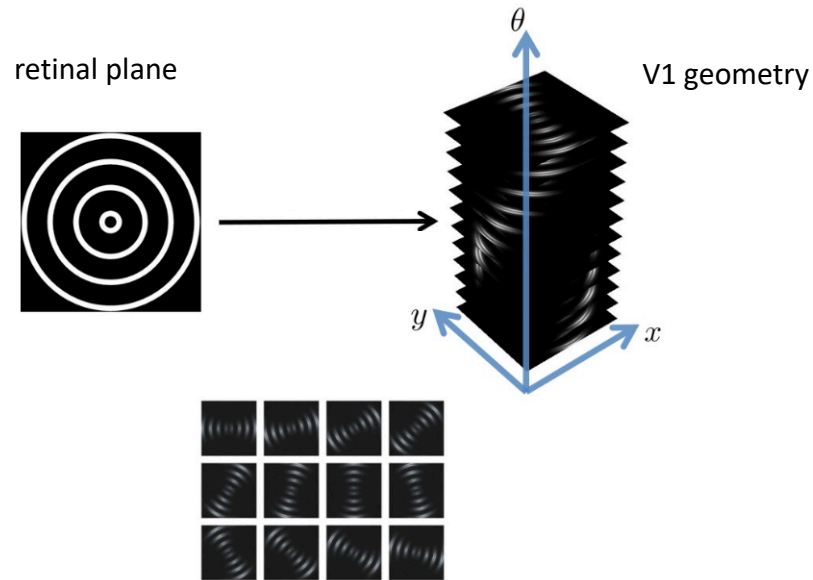
V1 geometry



- Output responses are obtained via filtering of the input image with Gabor functions: Gabor transform.
- Output response is a complex value since the wave content of $\Psi_{(x_0, y_0, \theta_0)}$ is a complex number.
- In general, we choose the real component of $O(x_0, y_0, \theta_0)$ to represent the output responses in V1 geometry.
- However, it is not the unique way: imaginary component, energy value ($\text{Re}^2(O) + \text{Im}^2(O)$) etc. are used as well in the literature.
- To come back to the retinal (image) plane, an inverse procedure should be applied for all $(x_0, y_0) \in M$:

$$I(x_0, y_0) = \int_{[0, \pi)} O(x_0, y_0, \theta_0) d\theta_0$$

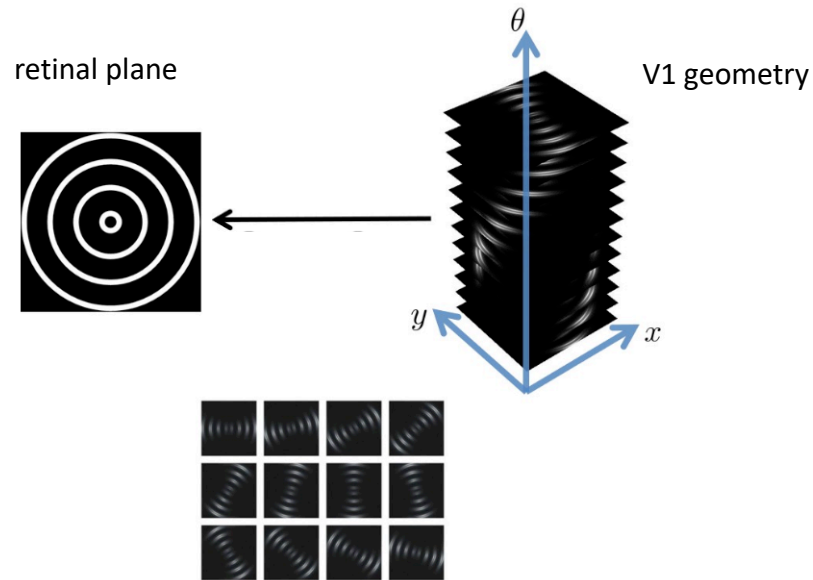
Citti-Sarti-Petitot model of the functional architecture



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Citti-Sarti-Petitot model of the functional architecture



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The end